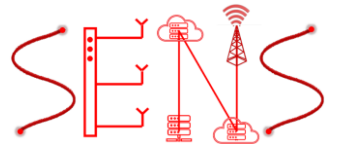


SAGE: A Real-Time AI System for Reducing Latency in NextG Cellular Networks

Aoyu Gong*, Raphael Cannatà*, Arman Maghsoudnia
Néstor Lomba Lomba, Dan Mihai Dumitriu, Haitham Hassanieh

Laboratory of Sensing and Networking Systems @ EPFL



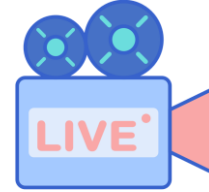
Next-Generation Cellular Networks



Autonomous Vehicles
(5 – 10 ms)



Rescue Services
(5 – 20 ms)



Live Production
(5 – 10 ms)



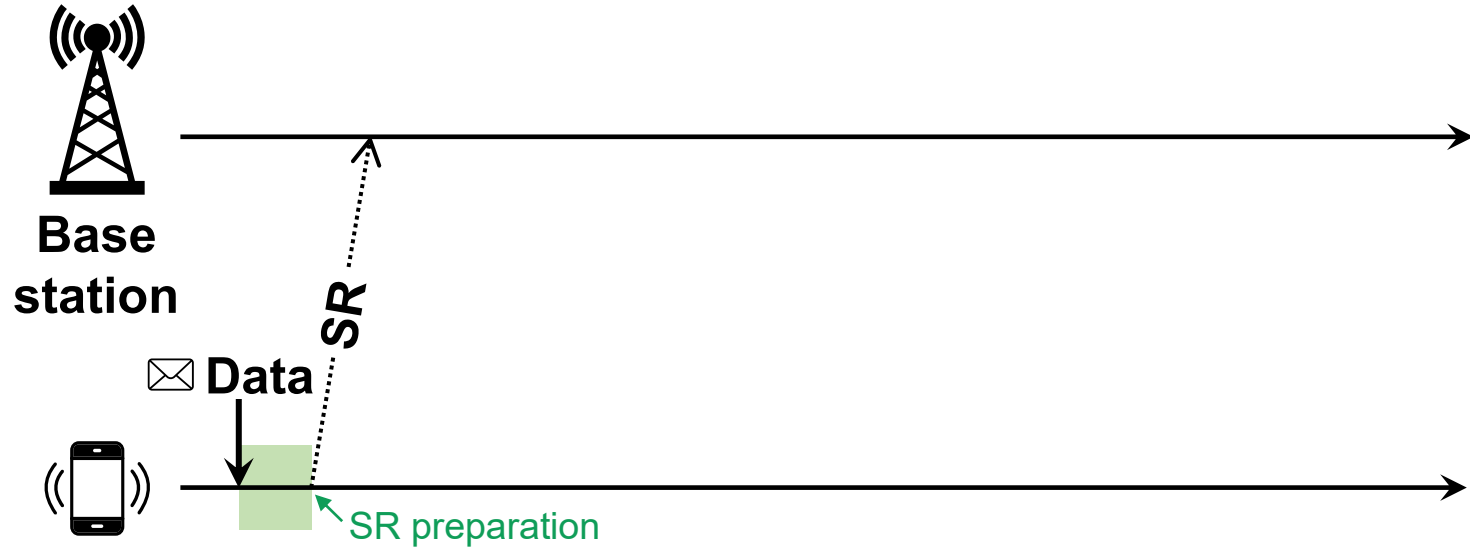
Industrial Automation
(1 – 2 ms)



Extended Reality
(4 – 7 ms)

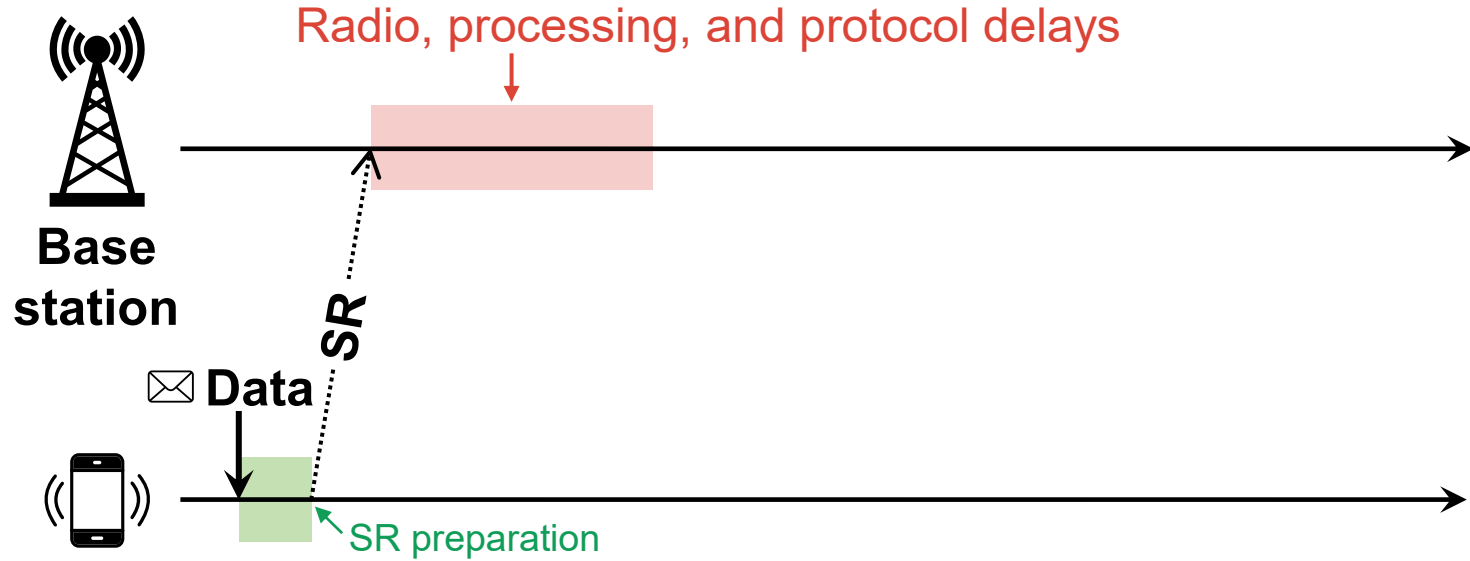
5G uplink scheduling is a key source of latency

Uplink Scheduling Bottlenecks



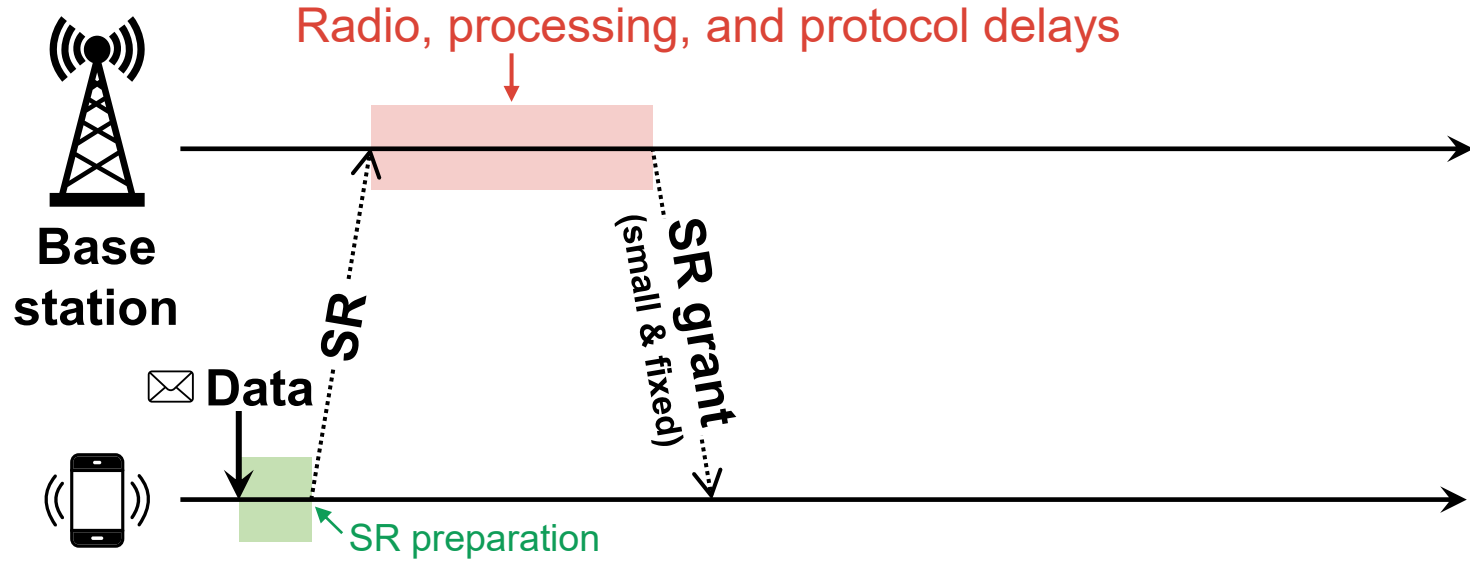
Scheduling Request (SR)

Uplink Scheduling Bottlenecks



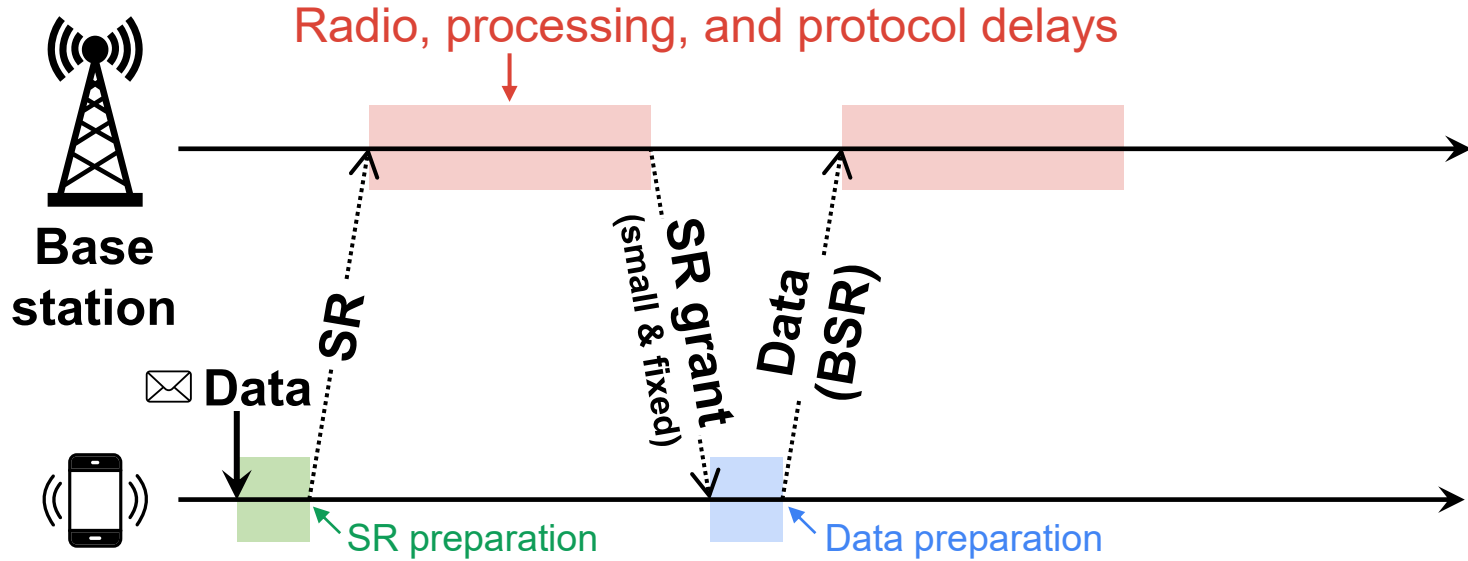
Multiple delays postpone the SR grant

Uplink Scheduling Bottlenecks



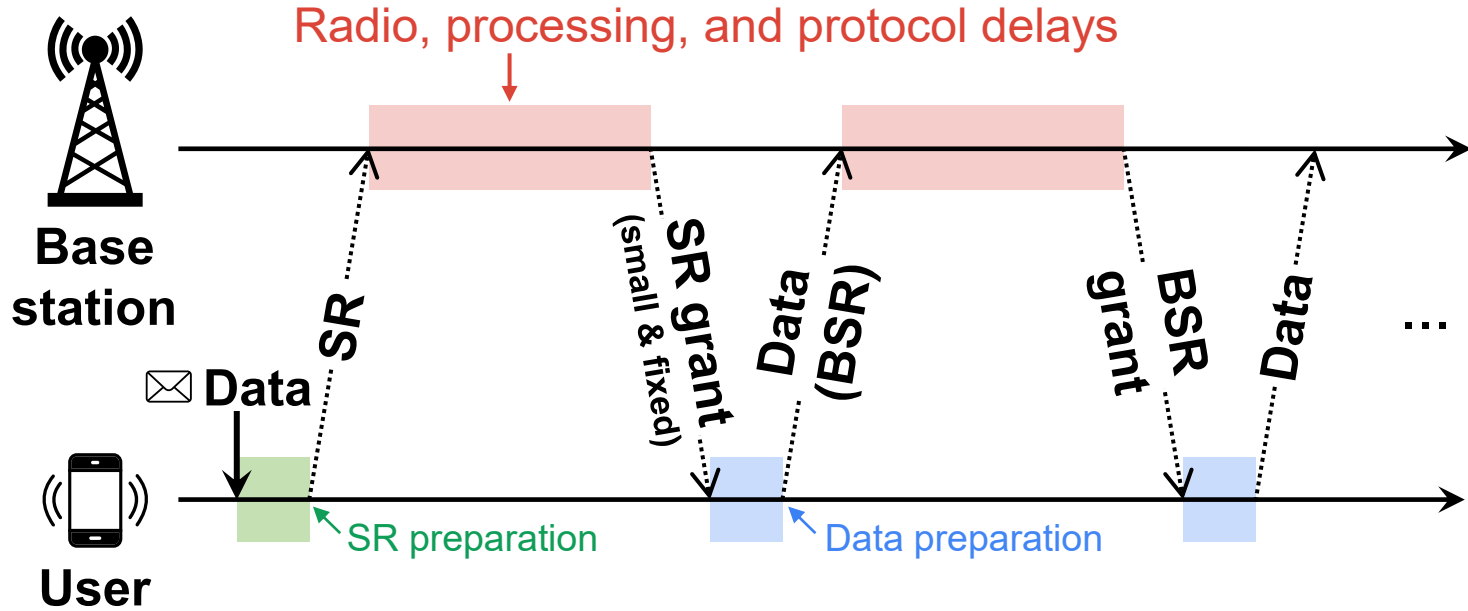
SR grant: small and fixed

Uplink Scheduling Bottlenecks



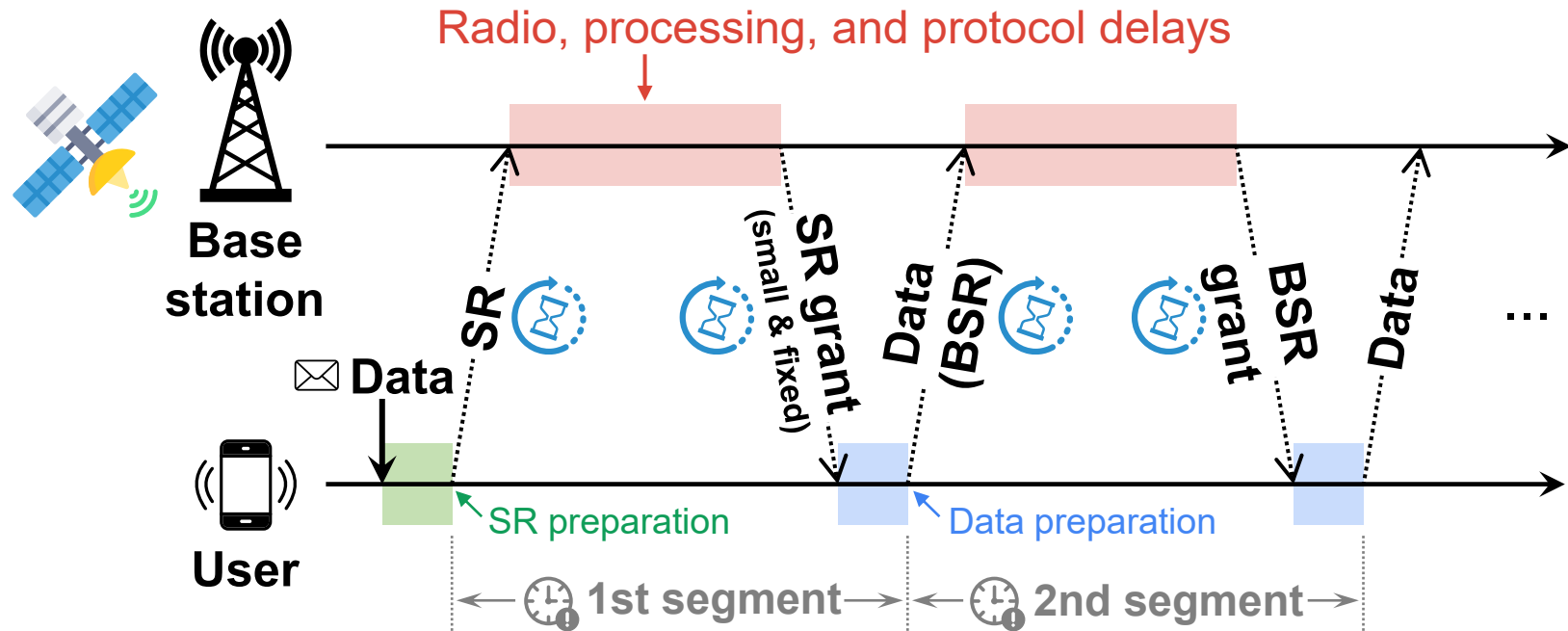
Buffer Status Report (BSR)

Uplink Scheduling Bottlenecks



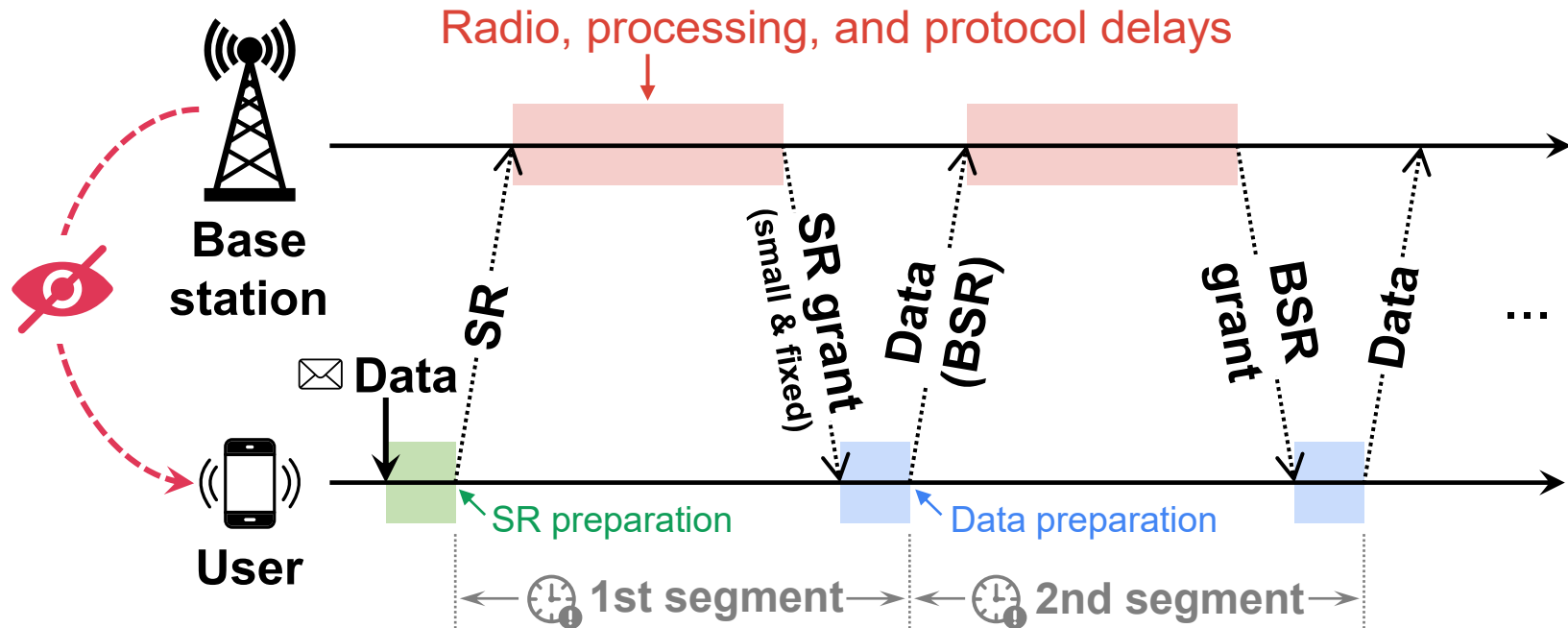
Scheduling becomes buffer-aware

Uplink Scheduling Bottlenecks



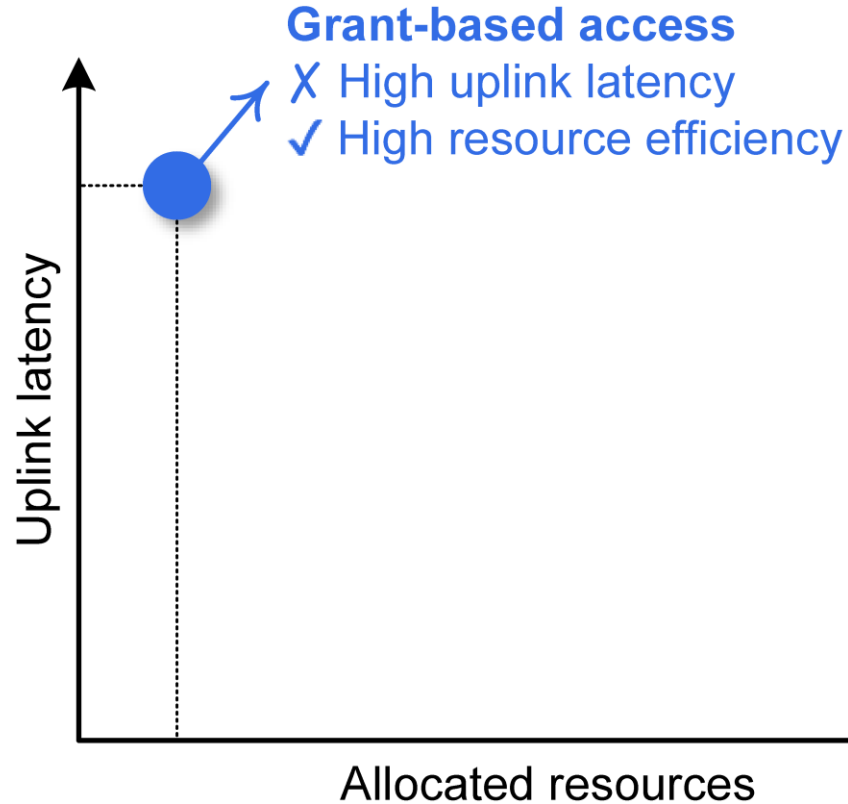
Two startup latency segments total 12 ms

Uplink Scheduling Bottlenecks

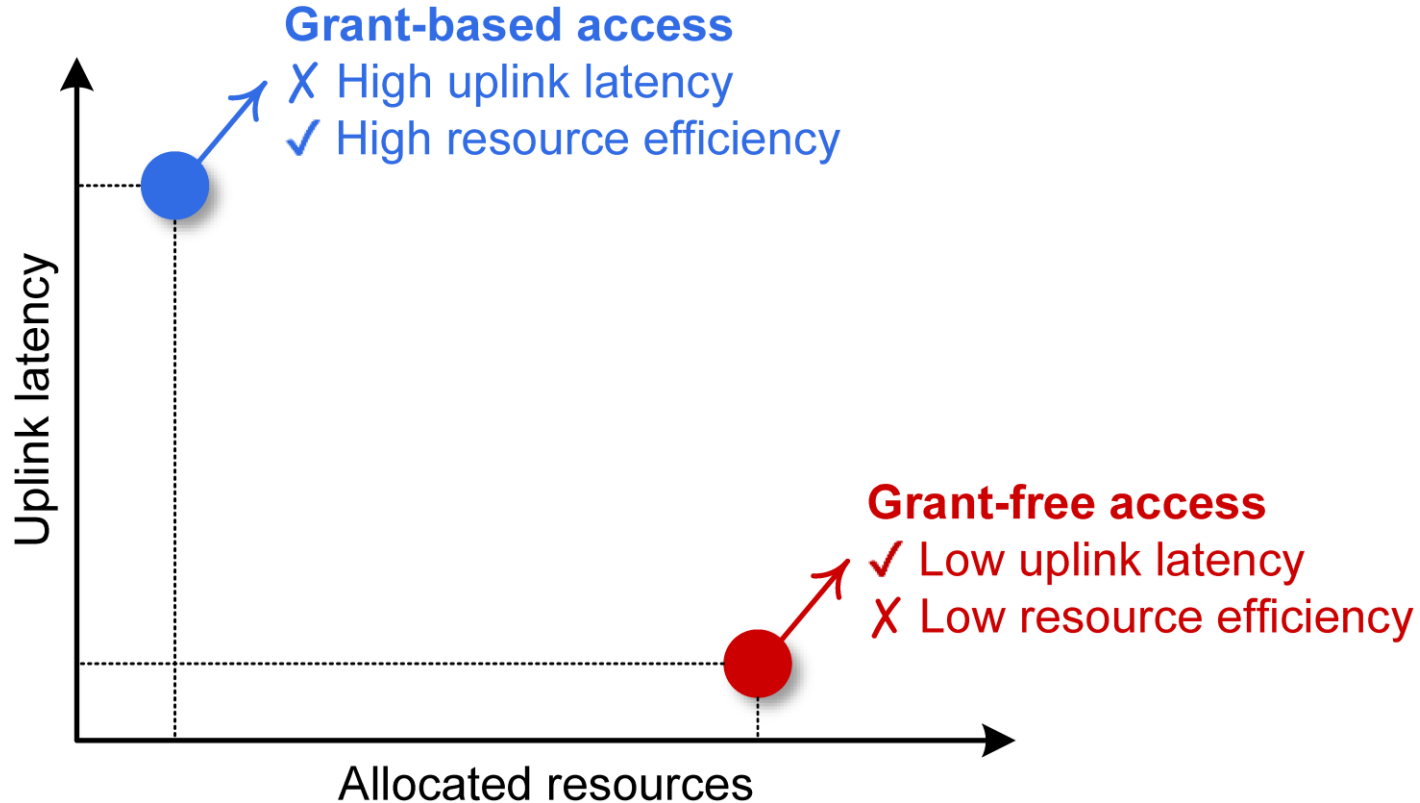


Base station has no knowledge of the user's buffer

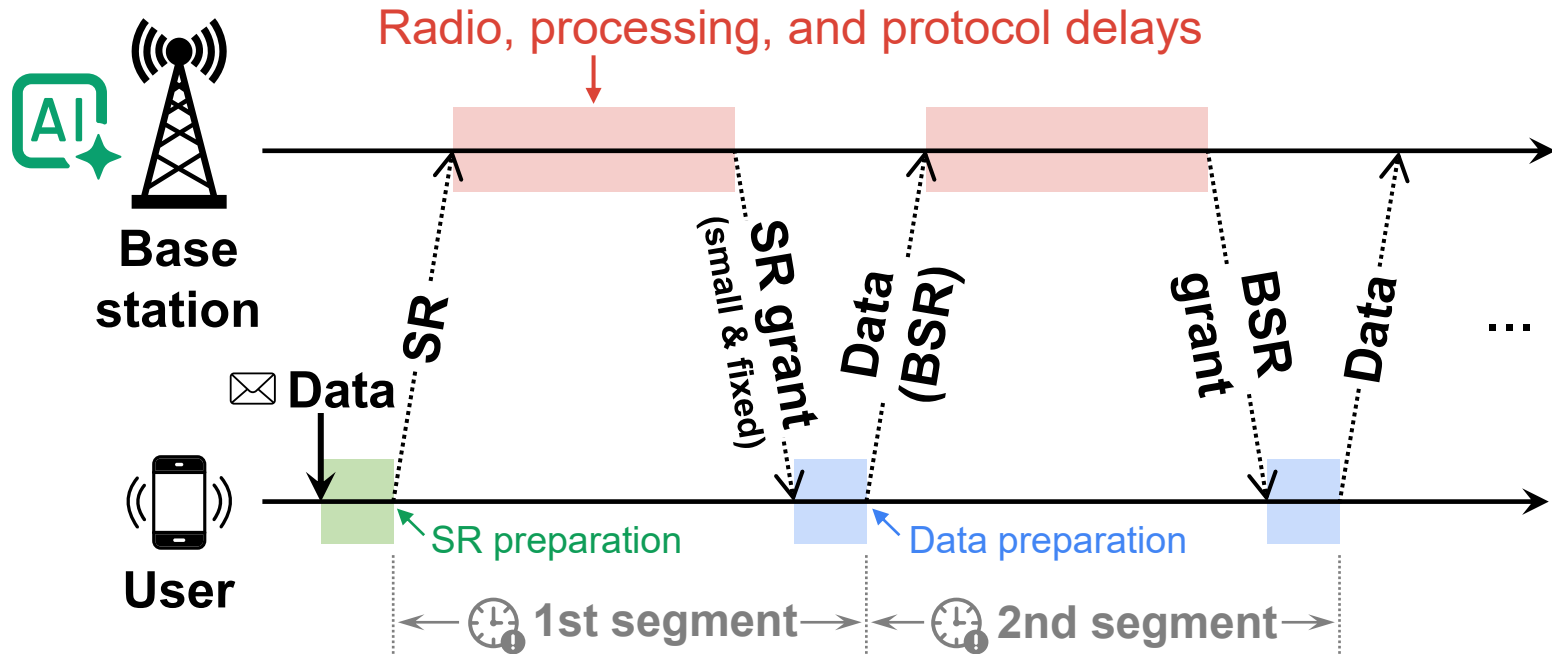
Grant-Based Access



Grant-Free Access

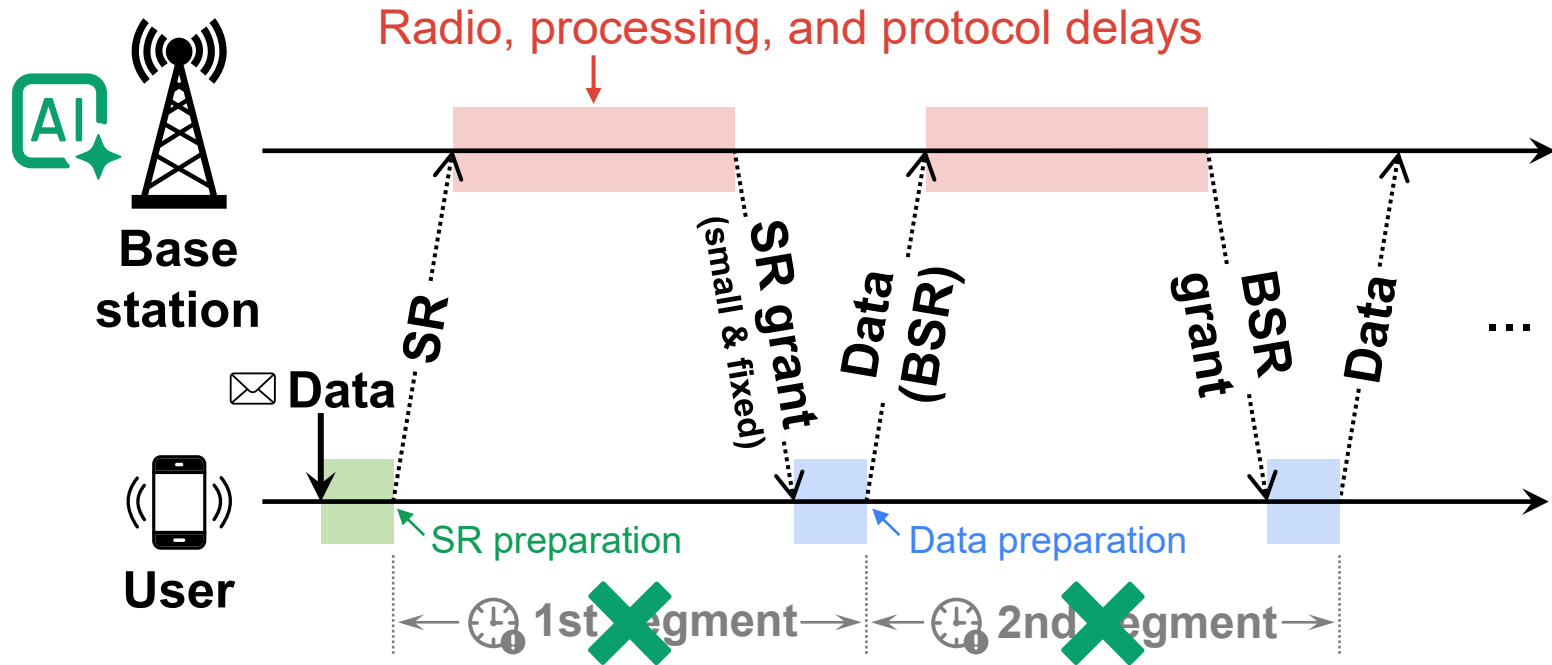


Our Idea



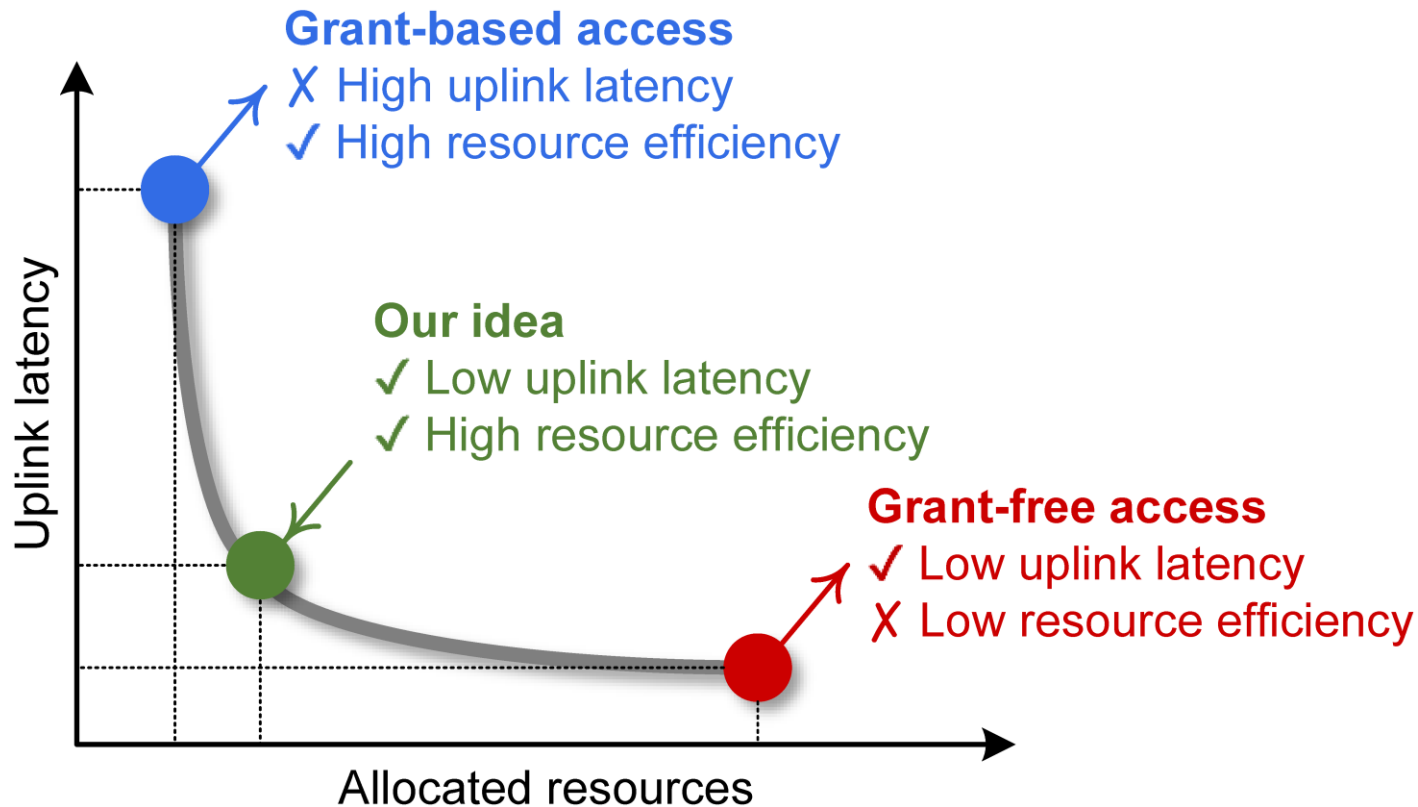
Predict future uplink data and proactively allocate resources

Our Idea

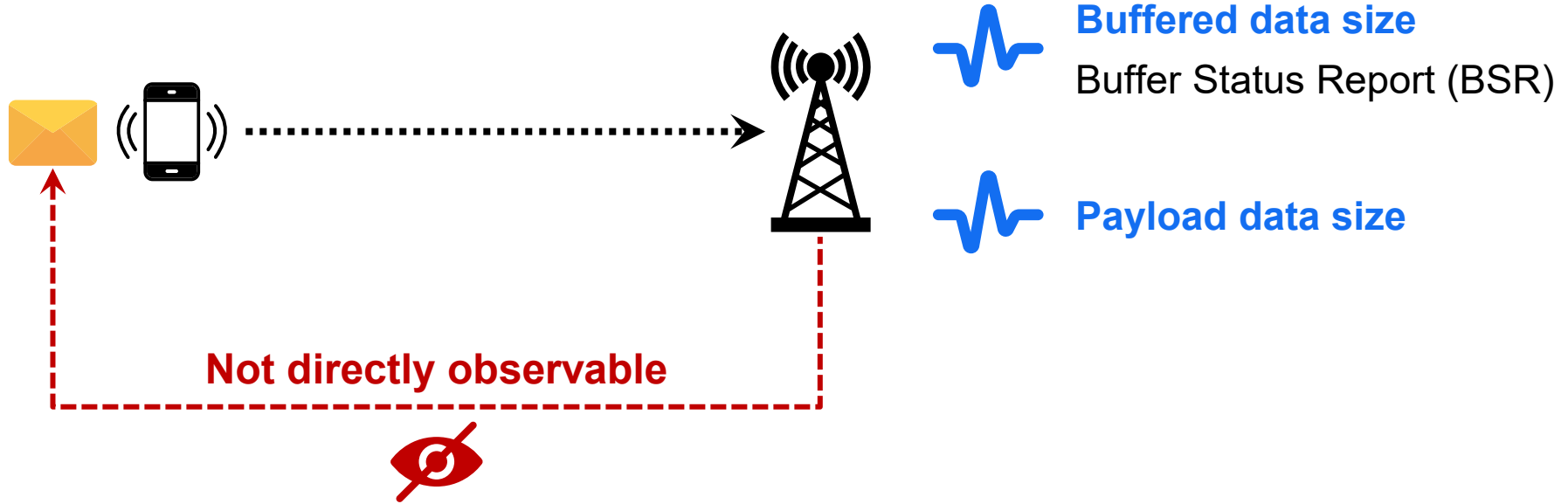


Predict future uplink data and proactively allocate resources

Our Idea

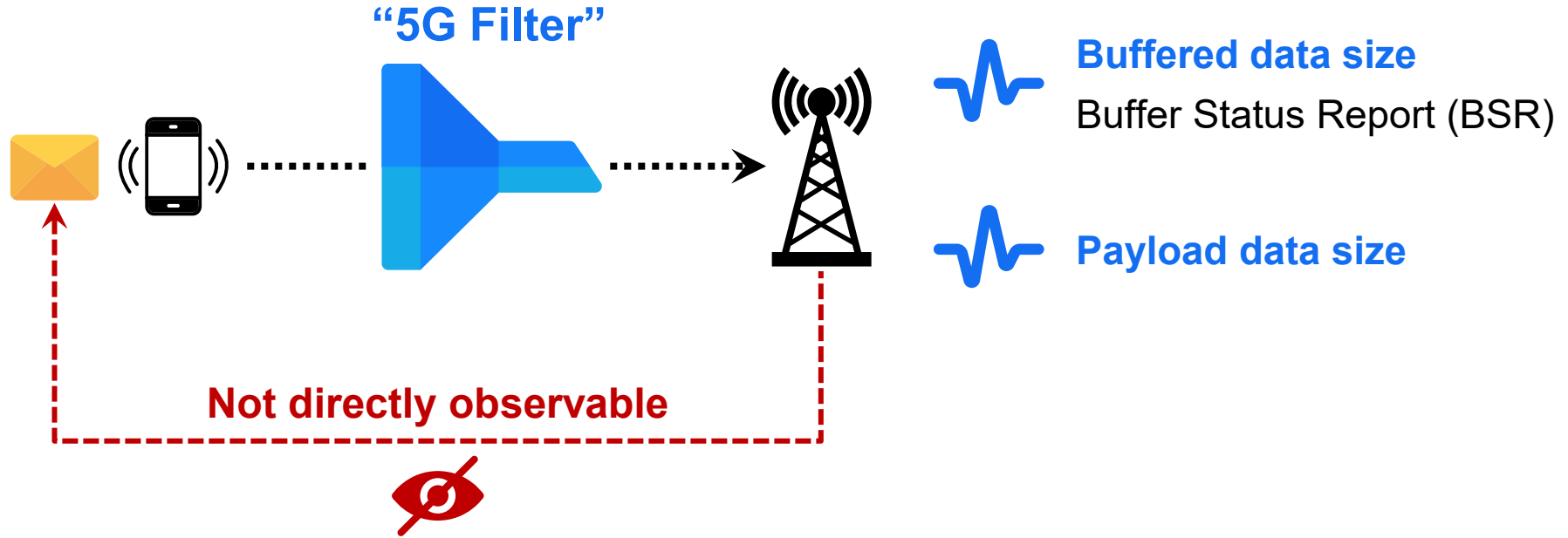


Challenges



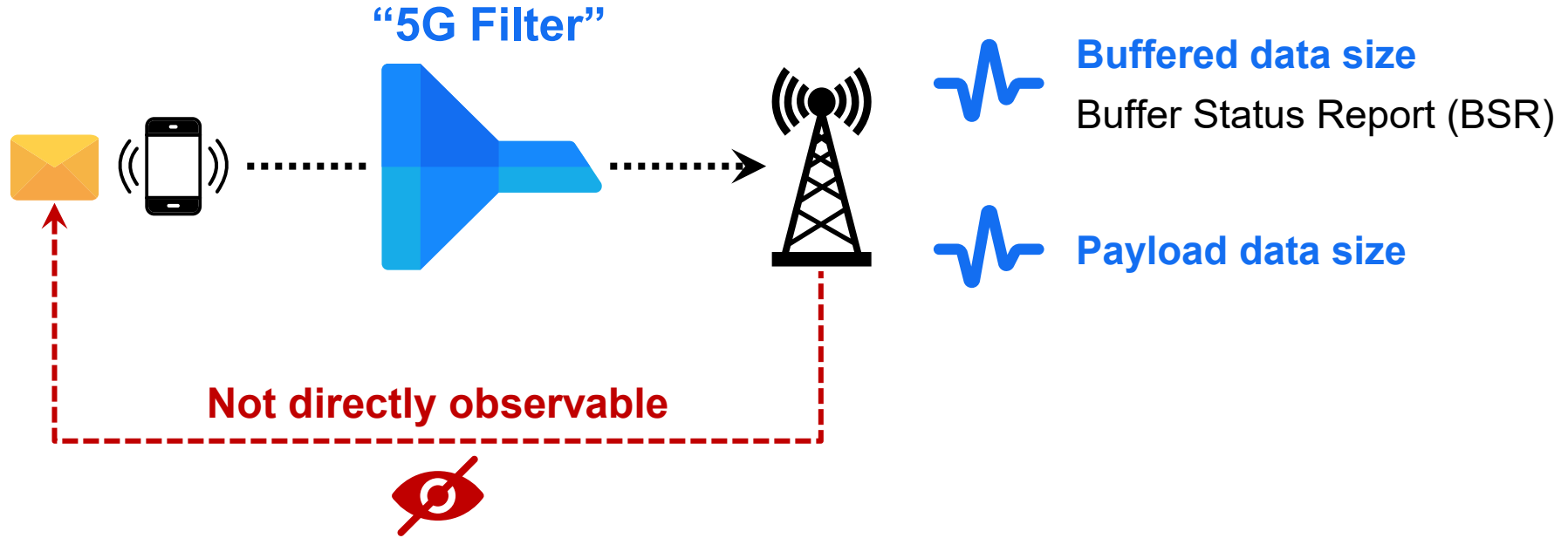
Base station **CANNOT** directly observe application traffic at the UE

Challenges



“What arrived at user” → **5G System** → “What base station sees”

Challenges

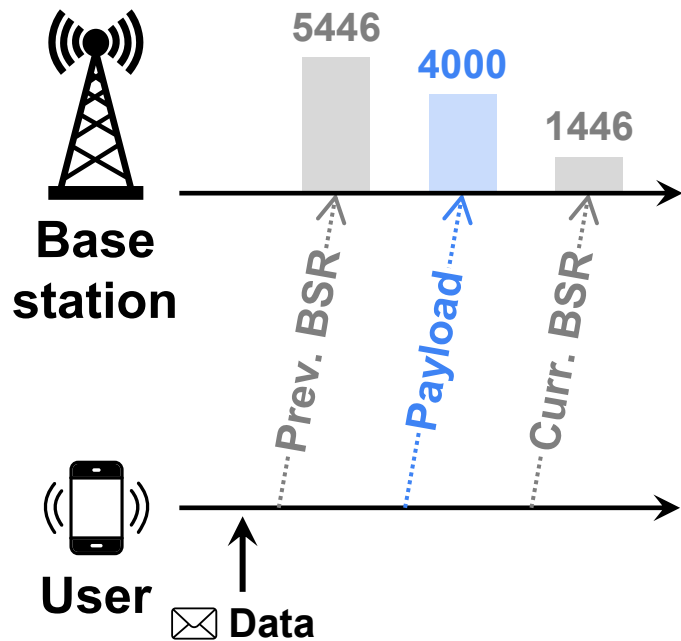


① System dynamics

② BSR quantization

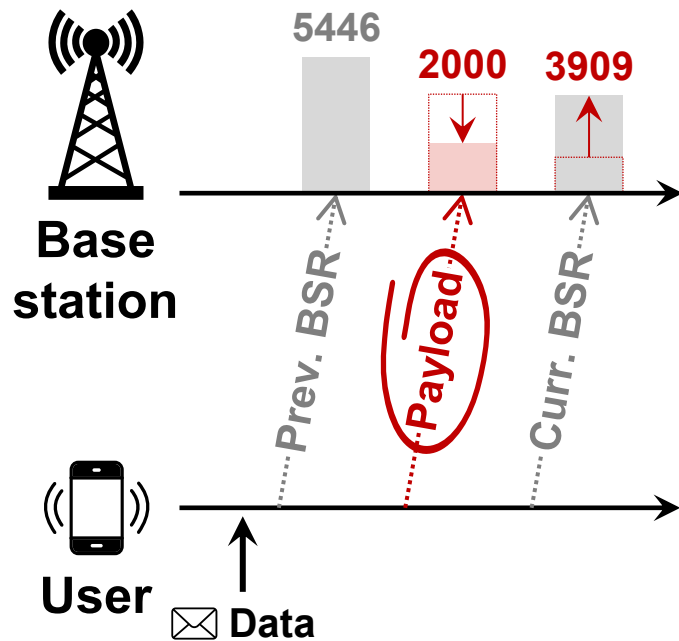
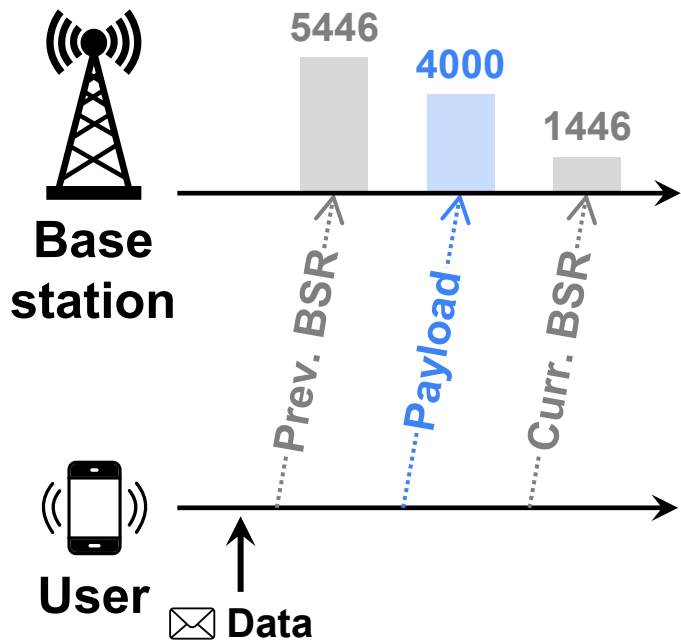
③ Packet loss

① System Dynamics



A single uplink burst → A sequence of BSRs and payloads

① System Dynamics

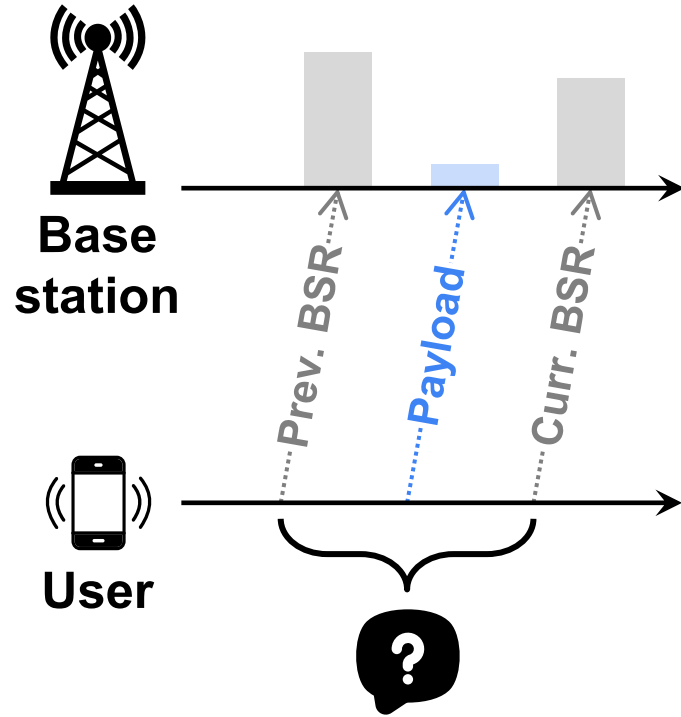


Affected by system dynamics



Generalizability

Estimated Traffic Arrivals

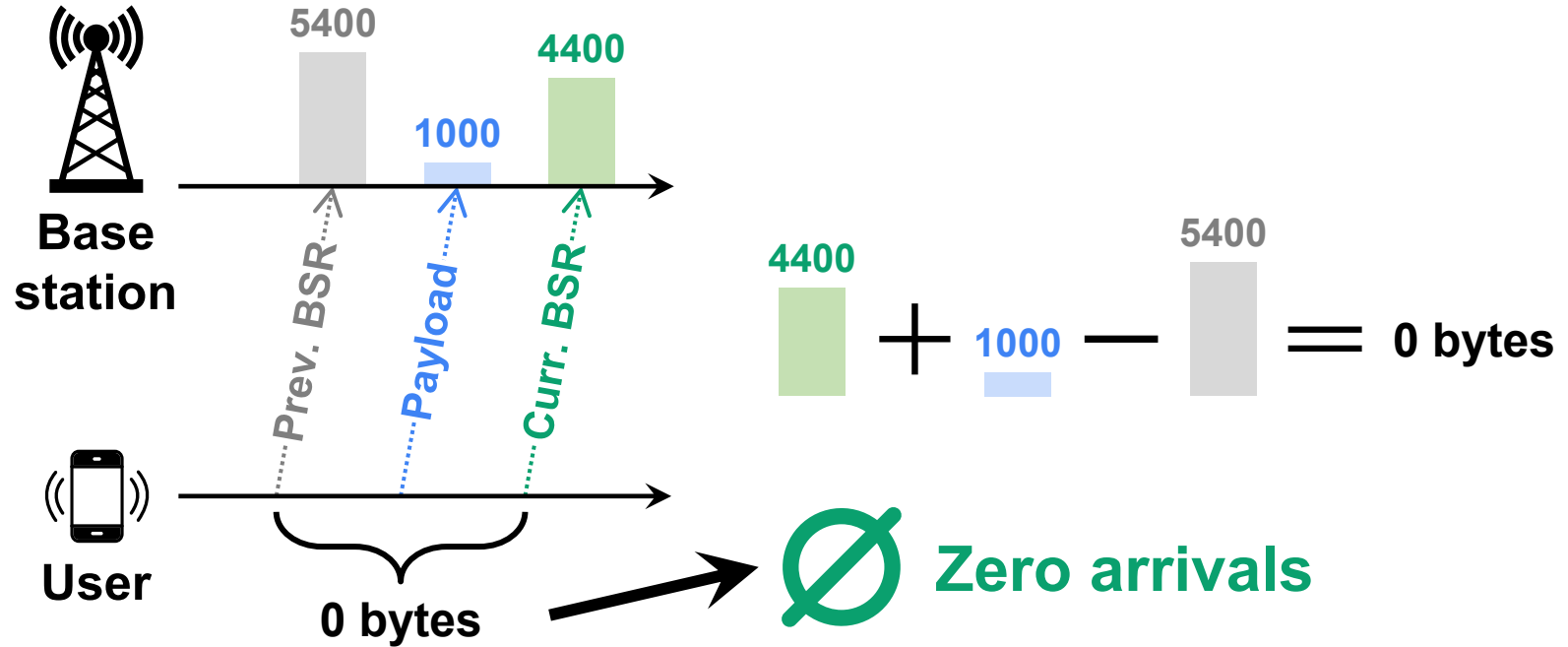


A visual equation showing a tall grey bar (Current BSR) followed by a plus sign, a short blue bar (Payload), a minus sign, and another tall grey bar (Previous BSR), followed by an equals sign and a speech bubble with a question mark.

$$\text{Current BSR} + \text{Payload} - \text{Previous BSR} = ?$$

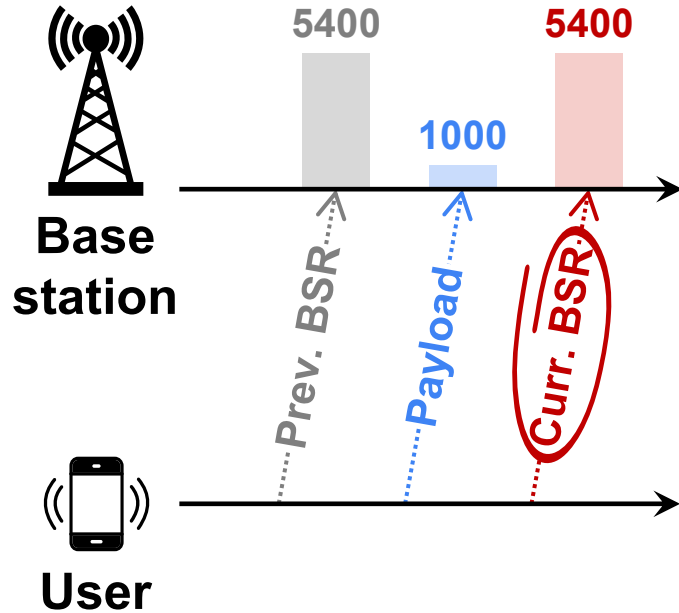
Estimated traffic arrival = Current BSR + Total payloads - Previous BSR

Estimated Traffic Arrivals



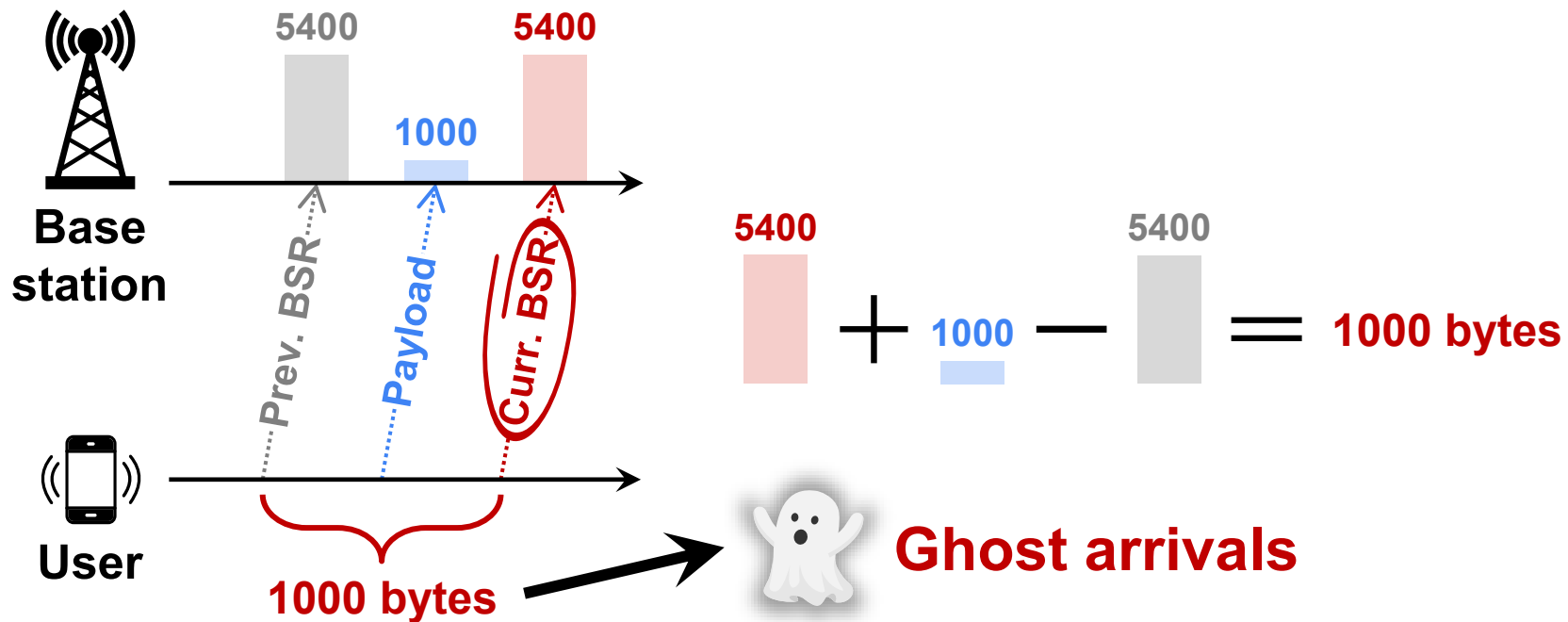
Estimated traffic arrival = Current BSR + Total payloads - Previous BSR

② BSR Quantization



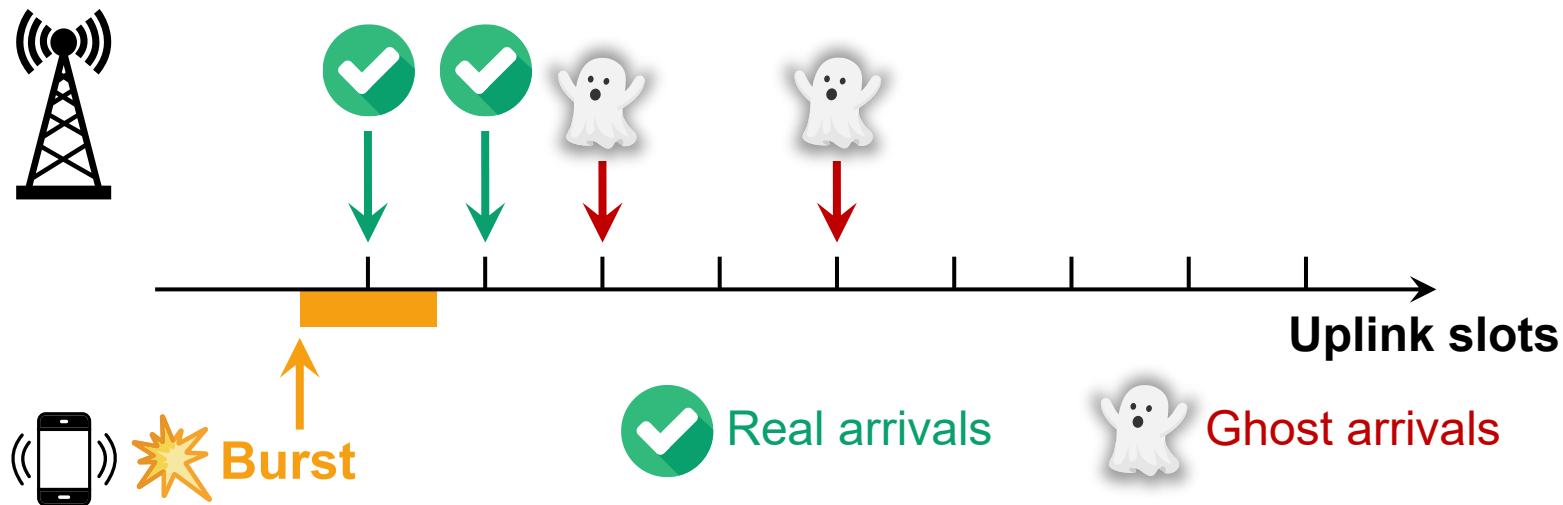
Estimated traffic arrival = Current BSR + Total payloads - Previous BSR

② BSR Quantization



Estimated traffic arrival = **Current BSR** + **Total payloads** - Previous BSR

“5G Filter” Distortion

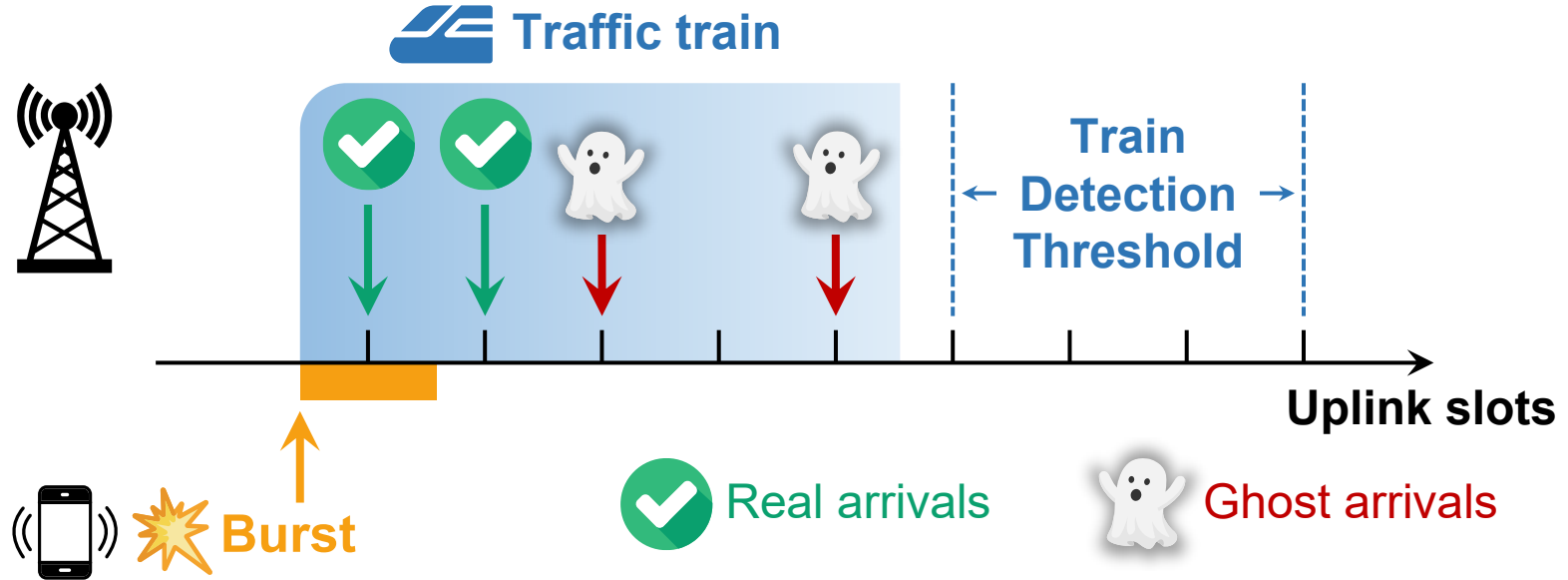


Distort inter-arrival statistics



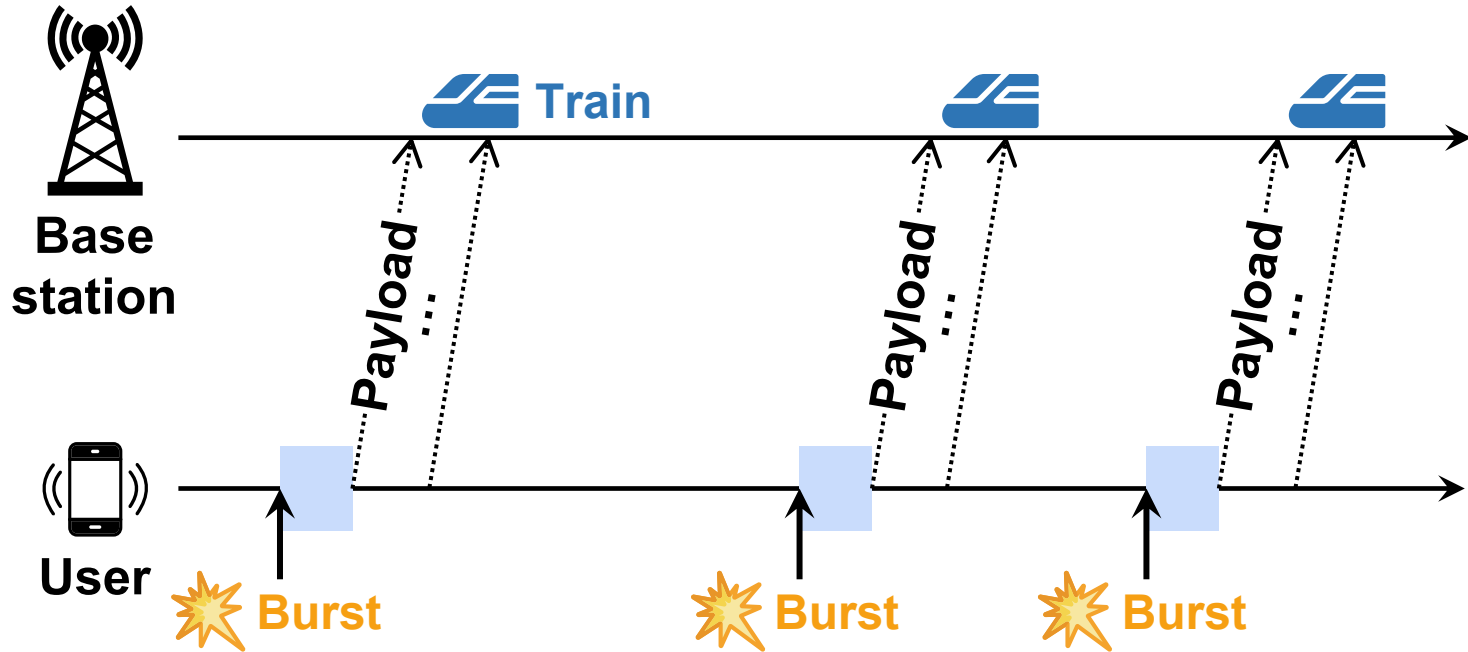
Inaccurate & inefficient

A New Abstraction: Traffic Train



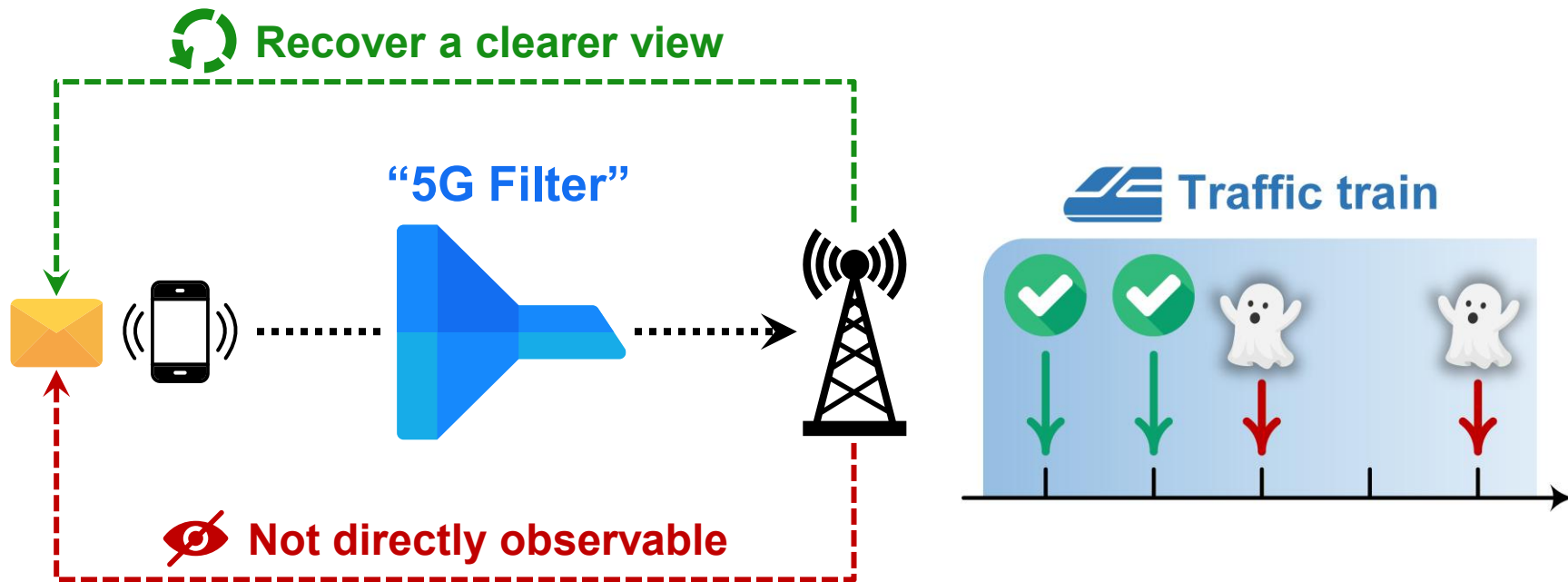
Group tightly spaced derived traffic arrivals into a traffic train

A New Abstraction: Traffic Train



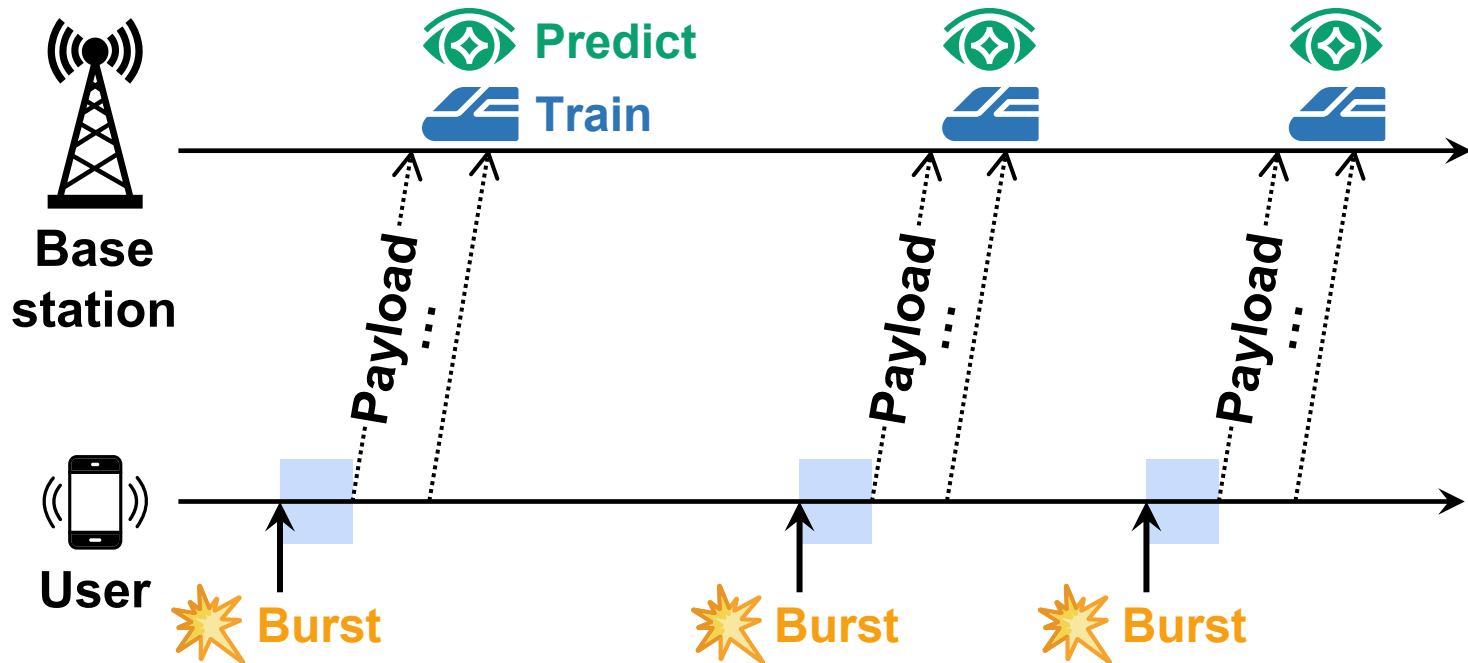
Map each uplink burst to a single traffic train

A New Abstraction: Traffic Train



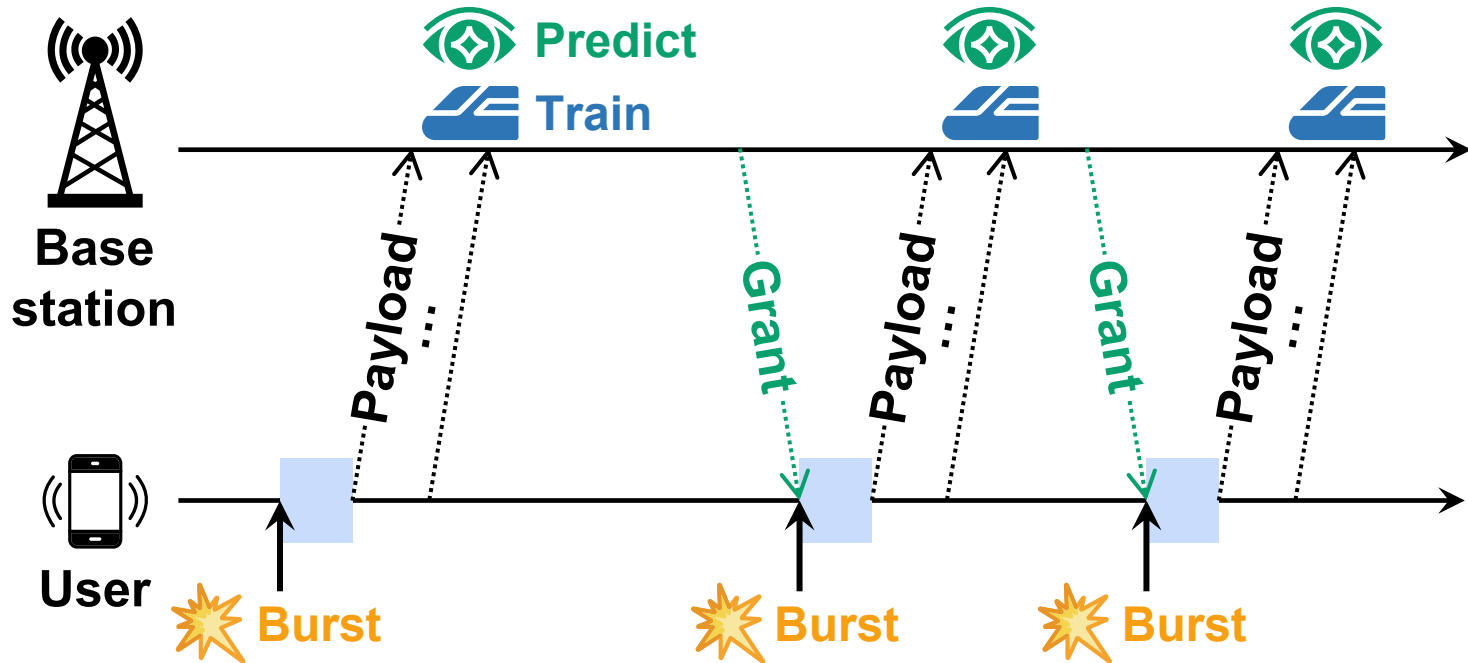
Recover a clearer view of application traffic

A New Abstraction: Traffic Train



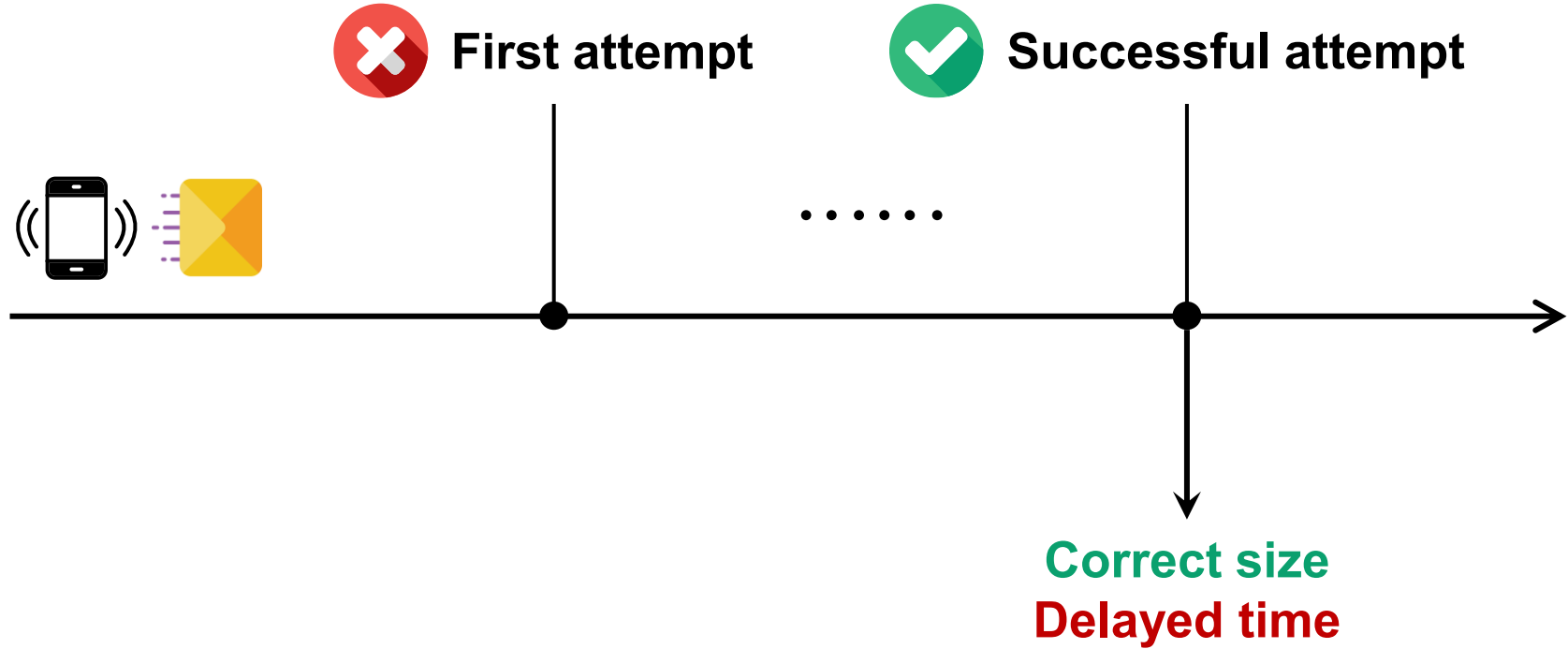
One prediction per traffic train (i.e., uplink burst)

A New Abstraction: Traffic Train



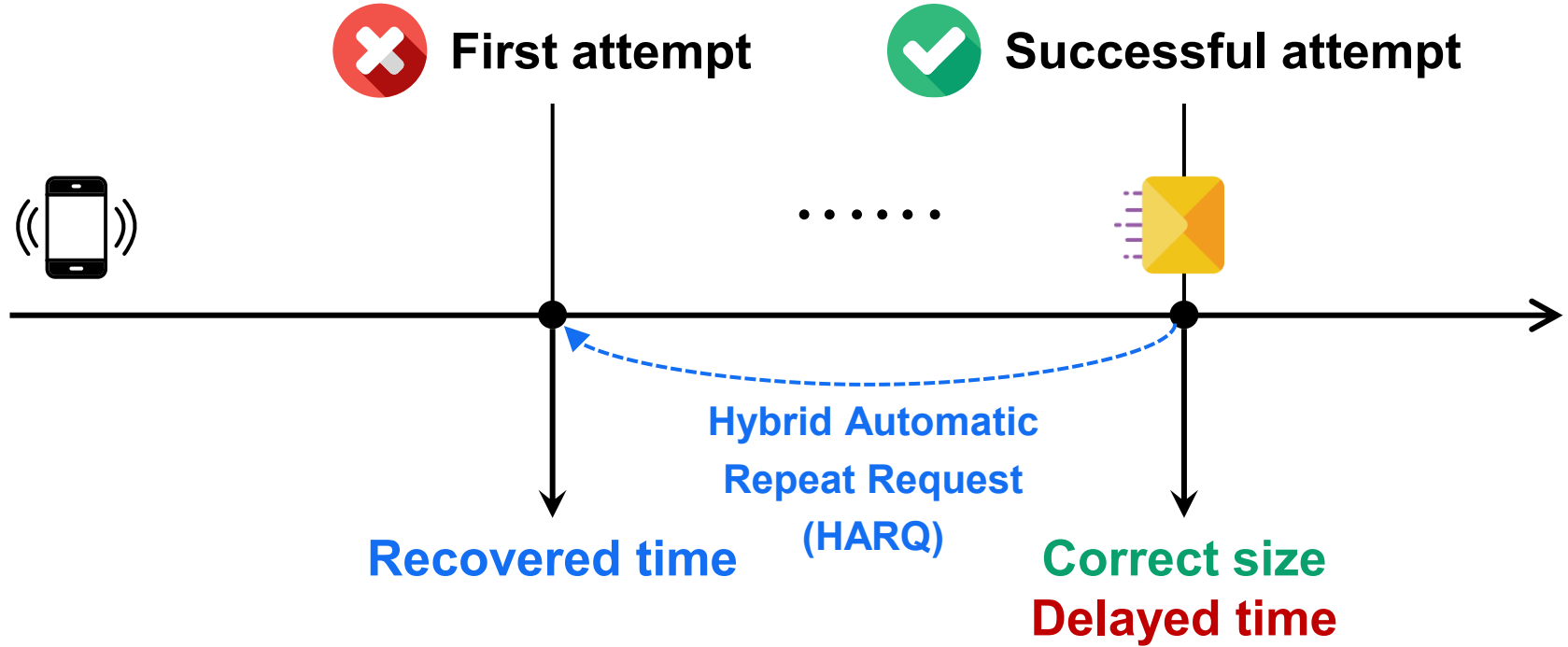
Predict the next traffic train and proactively allocate resources

③ Packet Loss



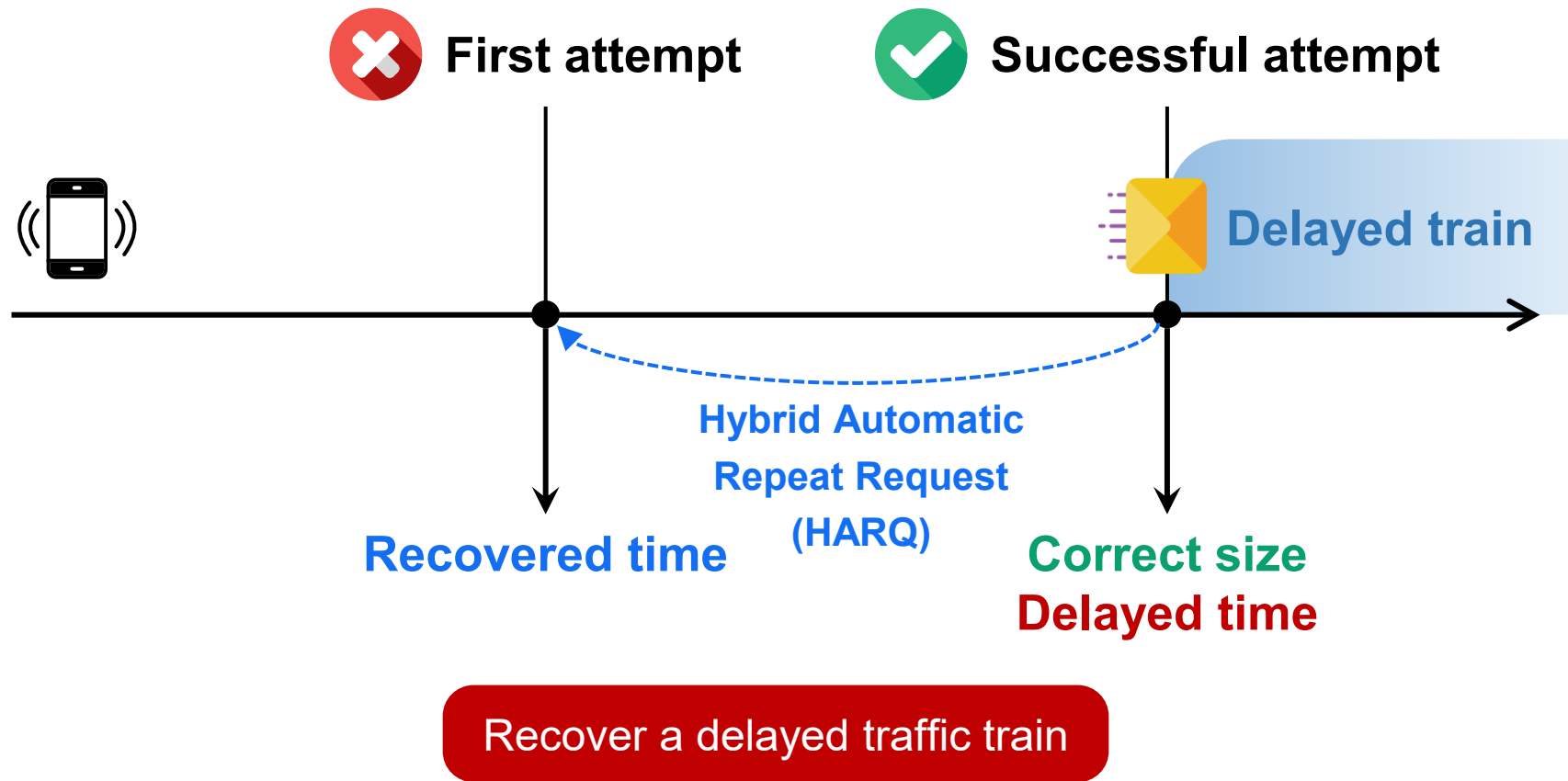
Retransmission delays the successful delivery of SDUs

③ Packet Loss

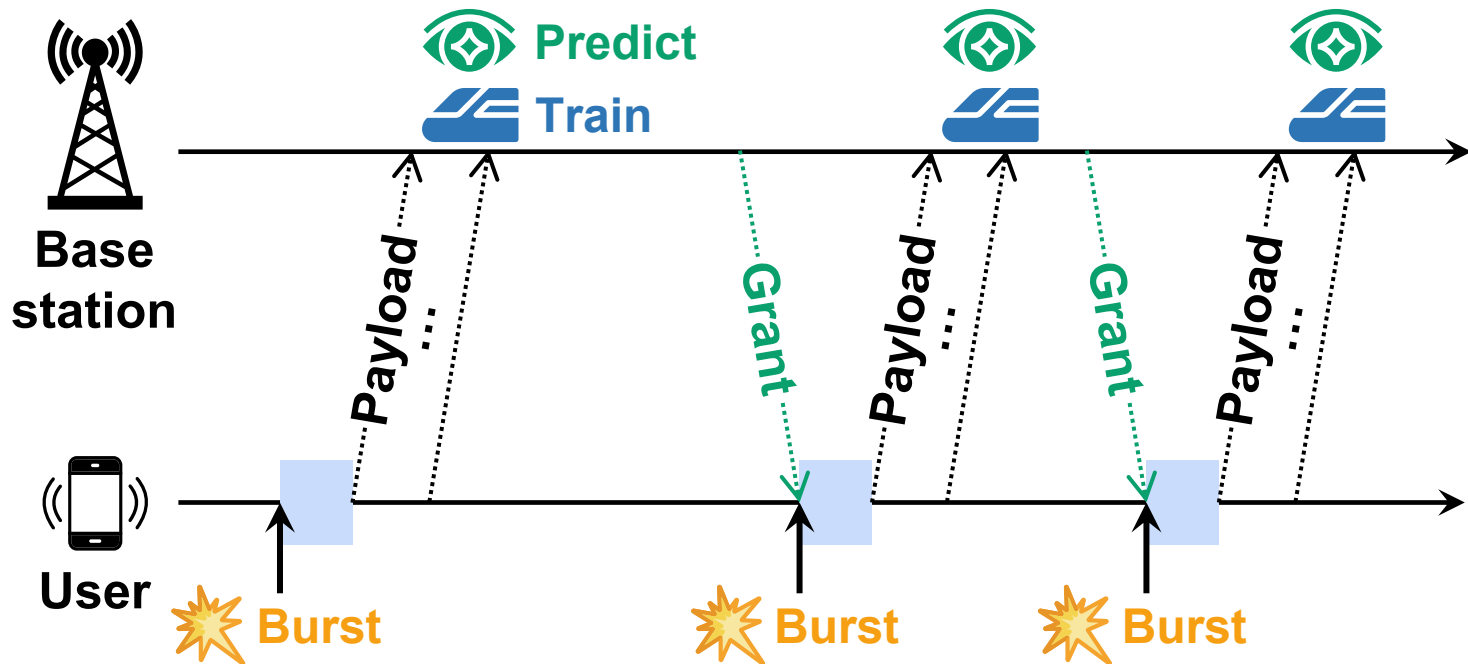


HARQ preserves the history of all transmission attempts

③ Packet Loss



A New Abstraction: Traffic Train



① System ~~×~~ dynamics

② BSR ~~×~~ quantization

③ Packet ~~×~~ loss

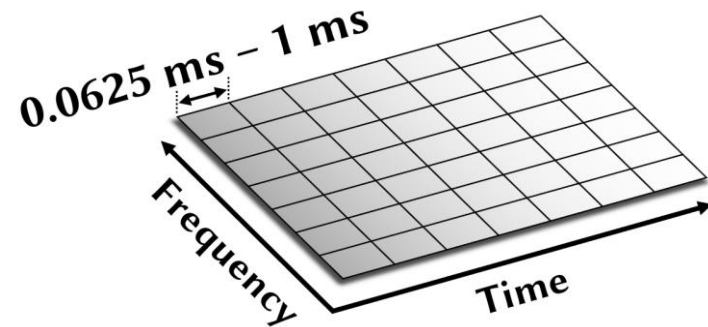
Challenges



Many emerging NextG applications

✗ Heavyweight, universal AI model

✓ Lightweight, expert AI models



Strict real-time constraints

Traffic-Aware Model Database

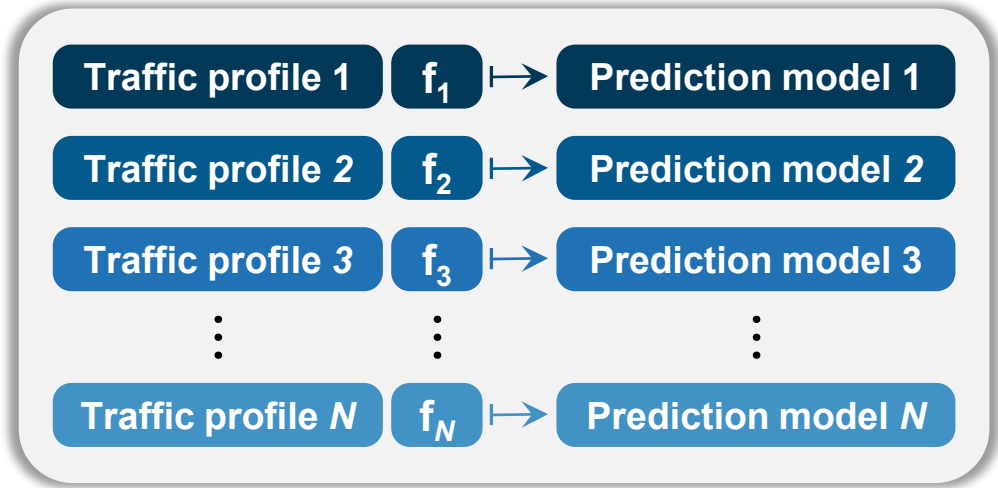


Traffic

f

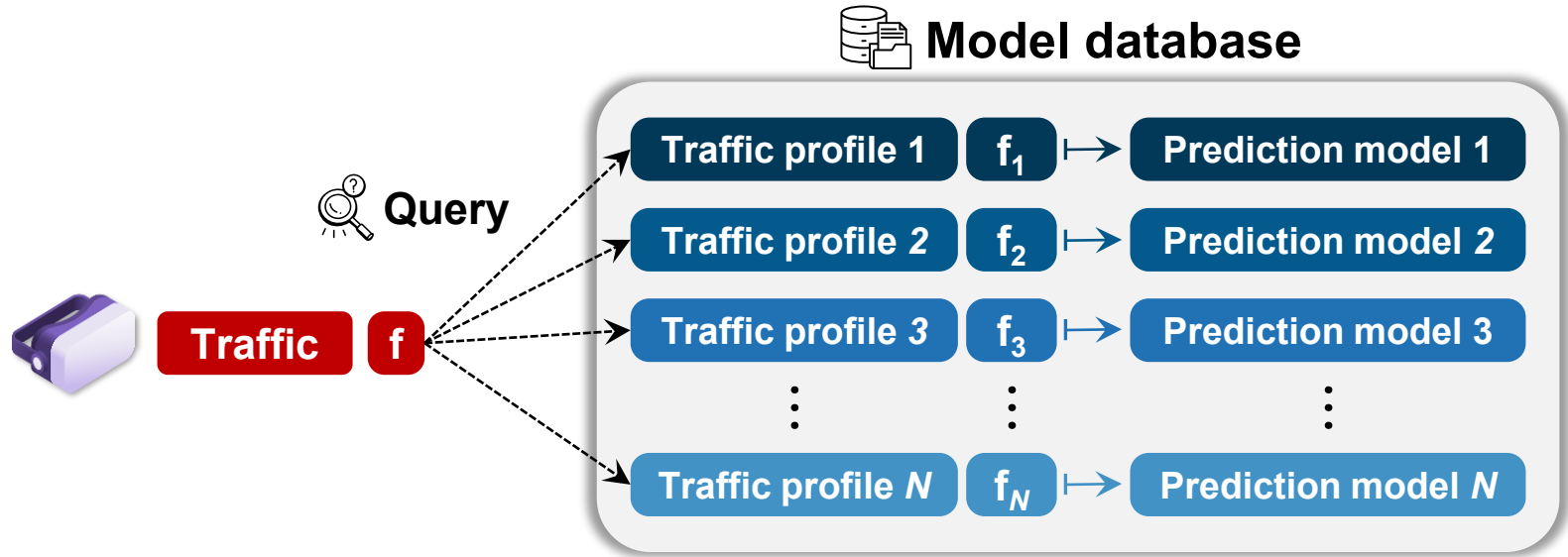


Model database



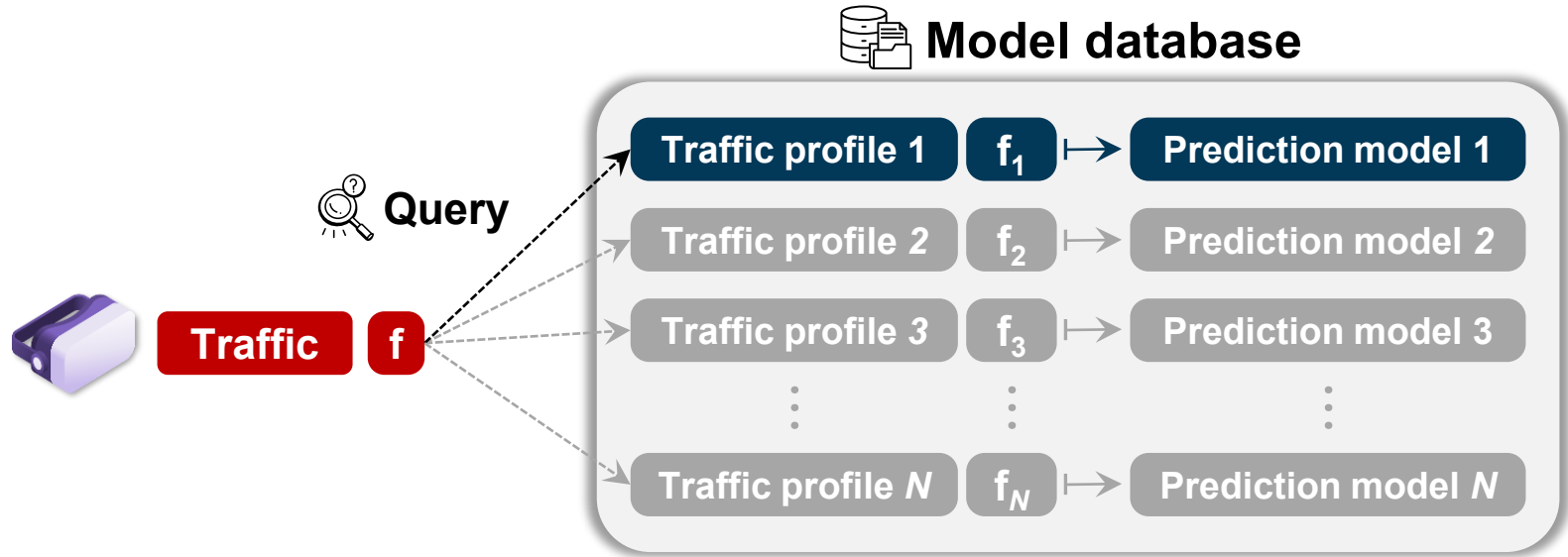
Construct a query vector using statistical features

Traffic-Aware Model Database



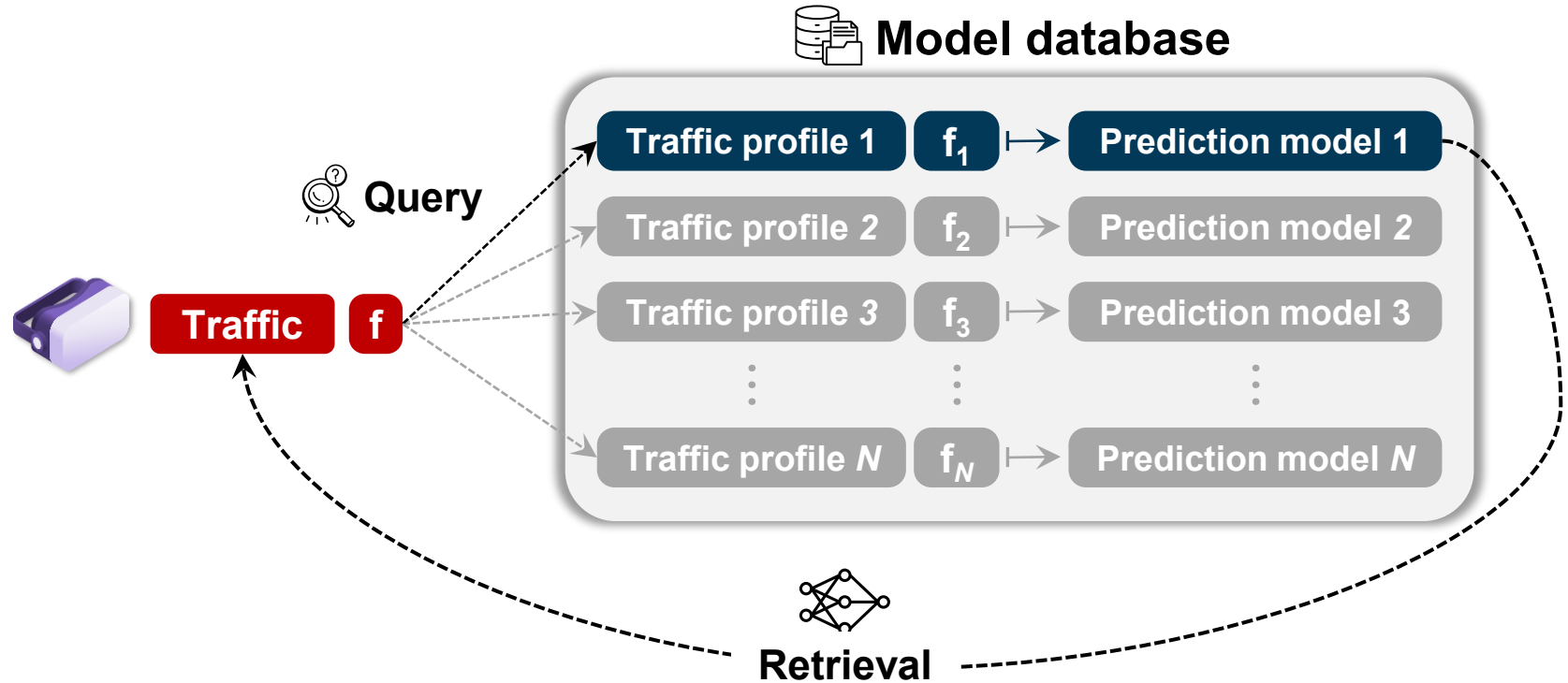
Measure the distance against all stored traffic profiles

Traffic-Aware Model Database



Measure the distance against all stored traffic profiles

Traffic-Aware Model Database

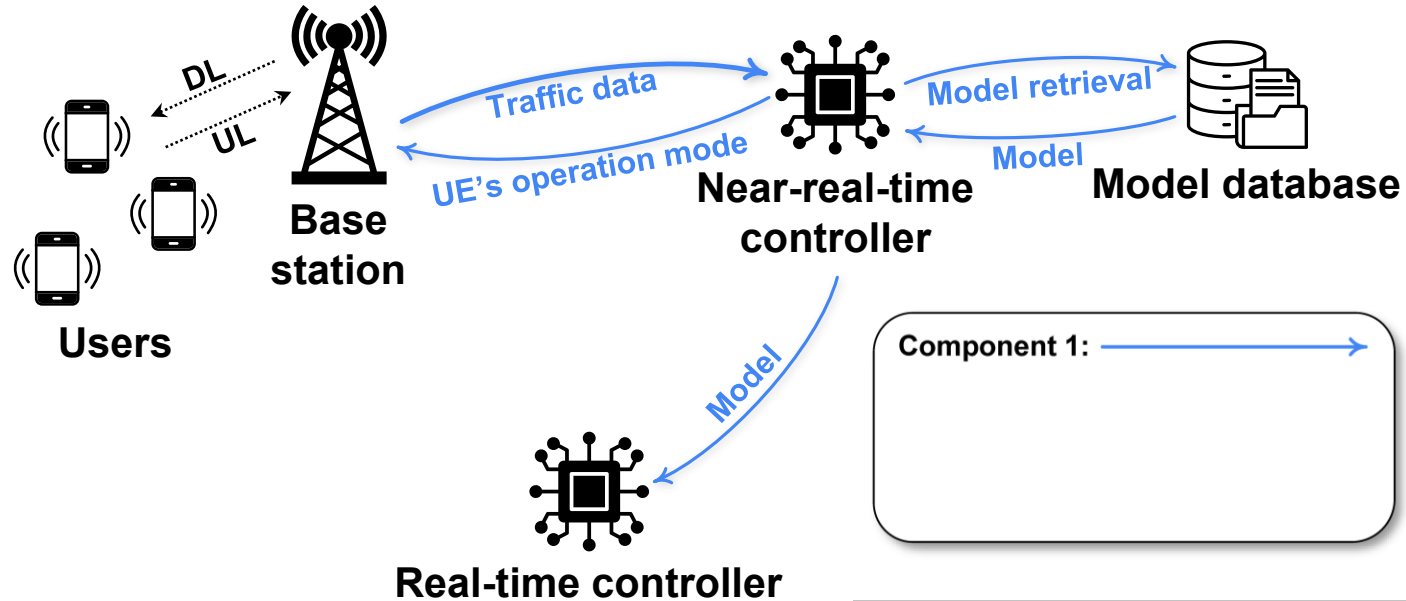


Plug-in design



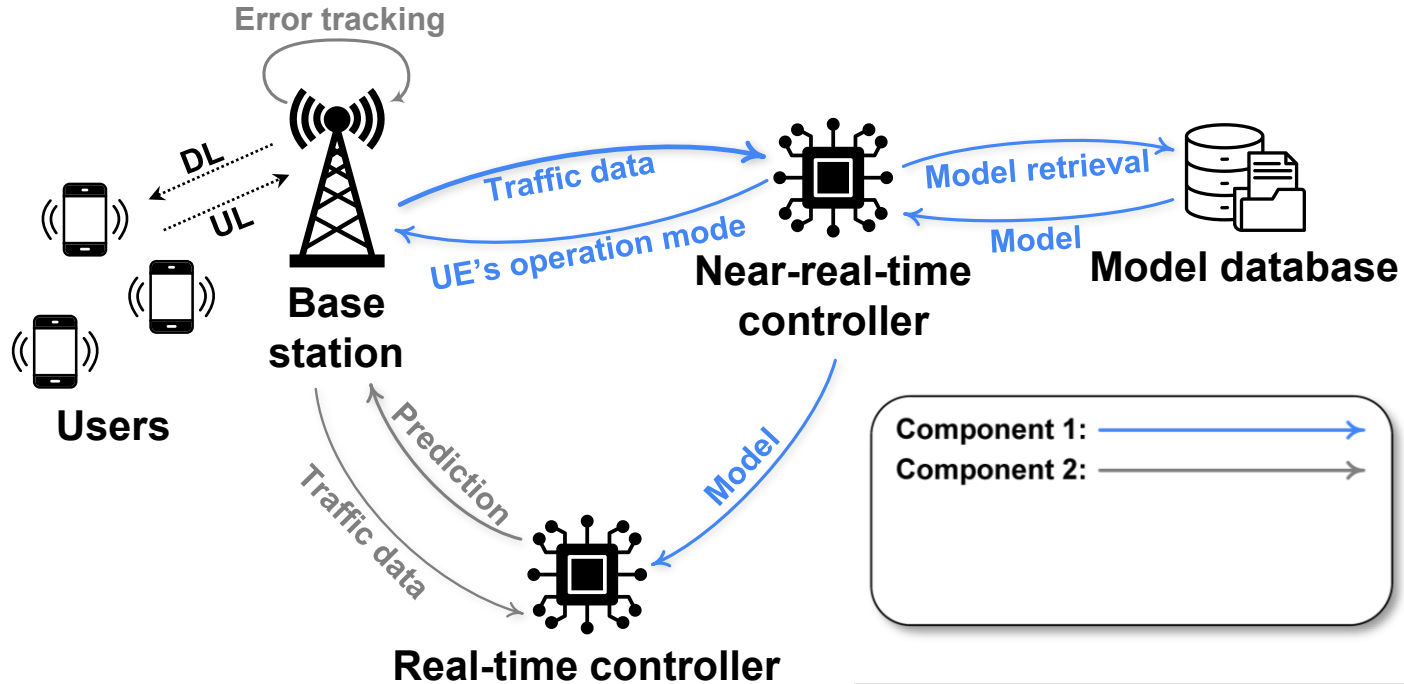
50 μ s on average

System Design: Traffic Characterization



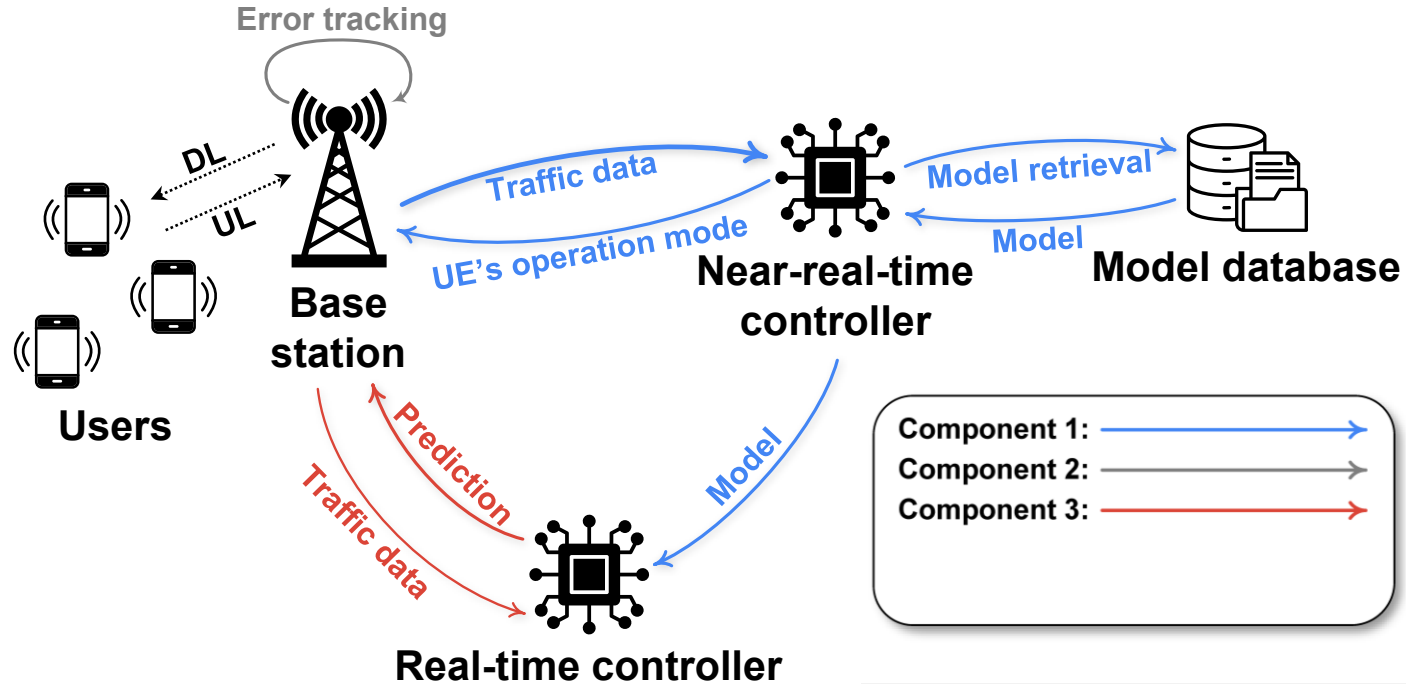
Characterize the traffic and retrieve the best-matching model

System Design: Model Verification



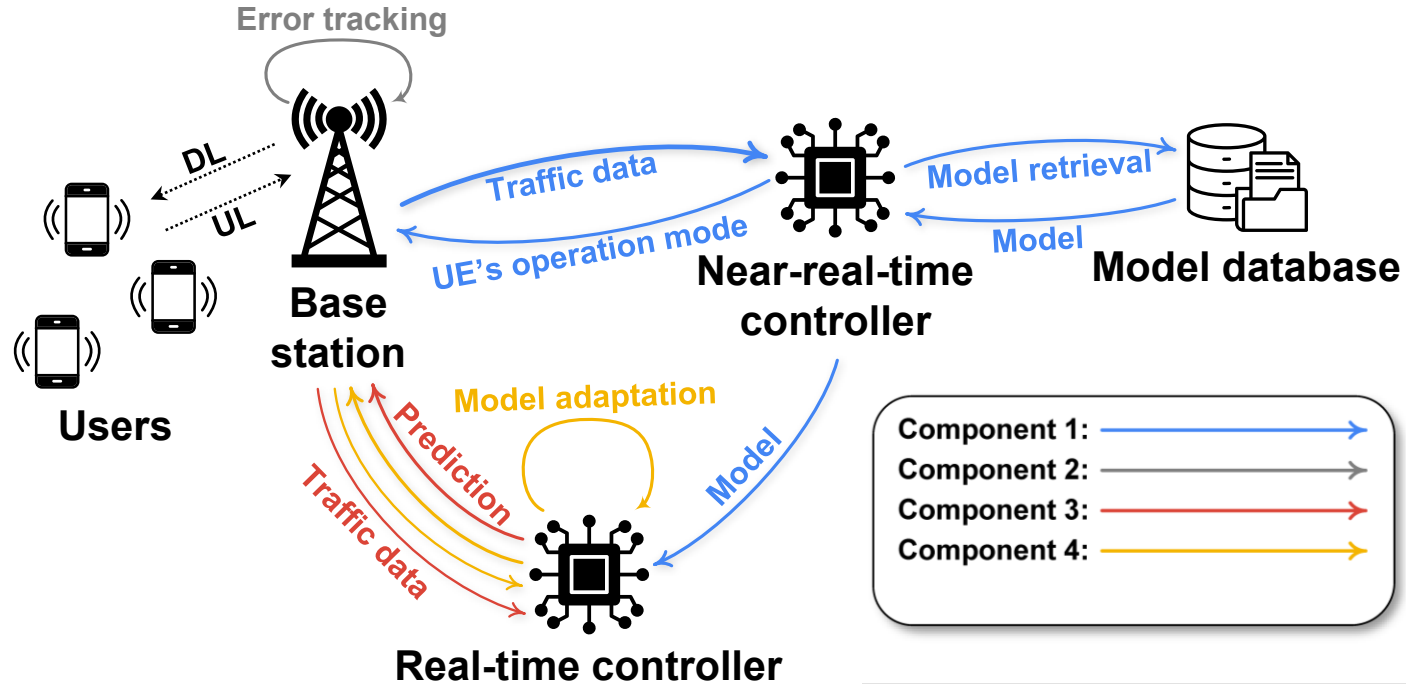
Verify the model's performance by tracking its prediction error

System Design: Prediction & Scheduling



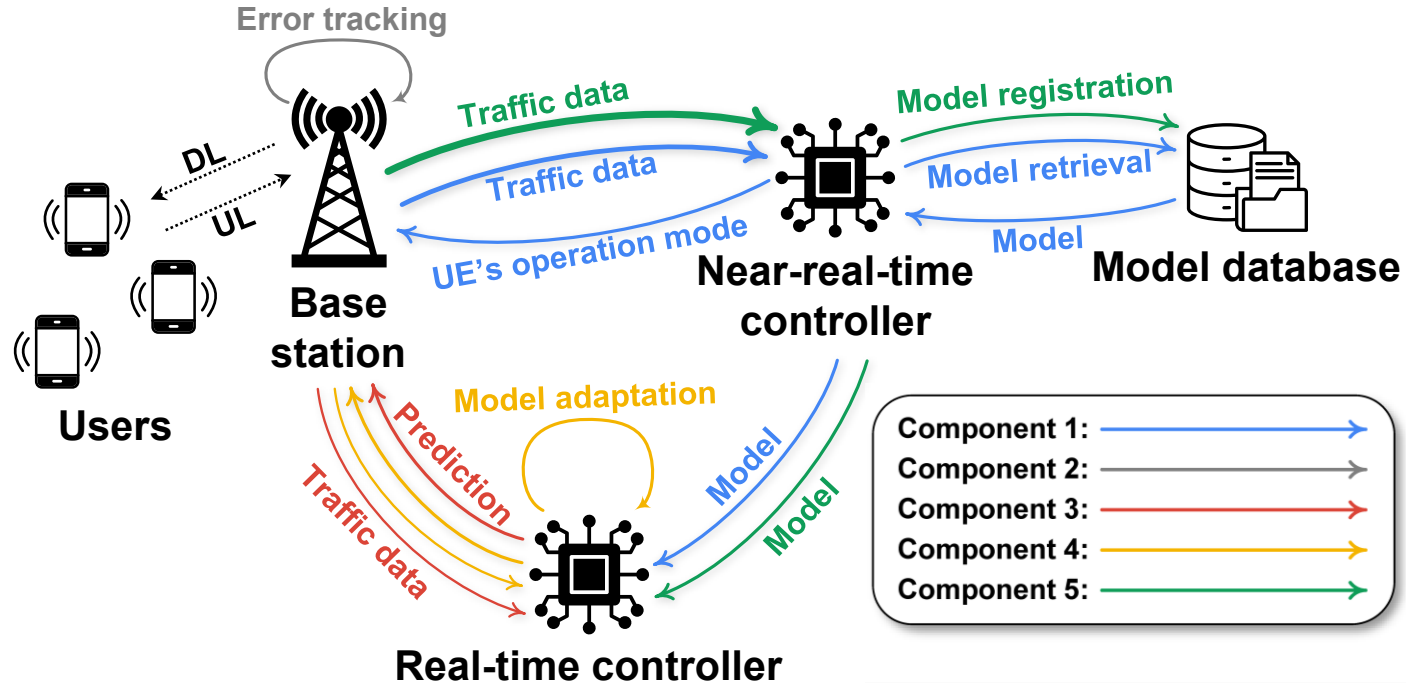
Allocate uplink resources proactively if the model performs well

System Design: Online Adaptation



Adapt the model online using continual learning

System Design: Database Updating



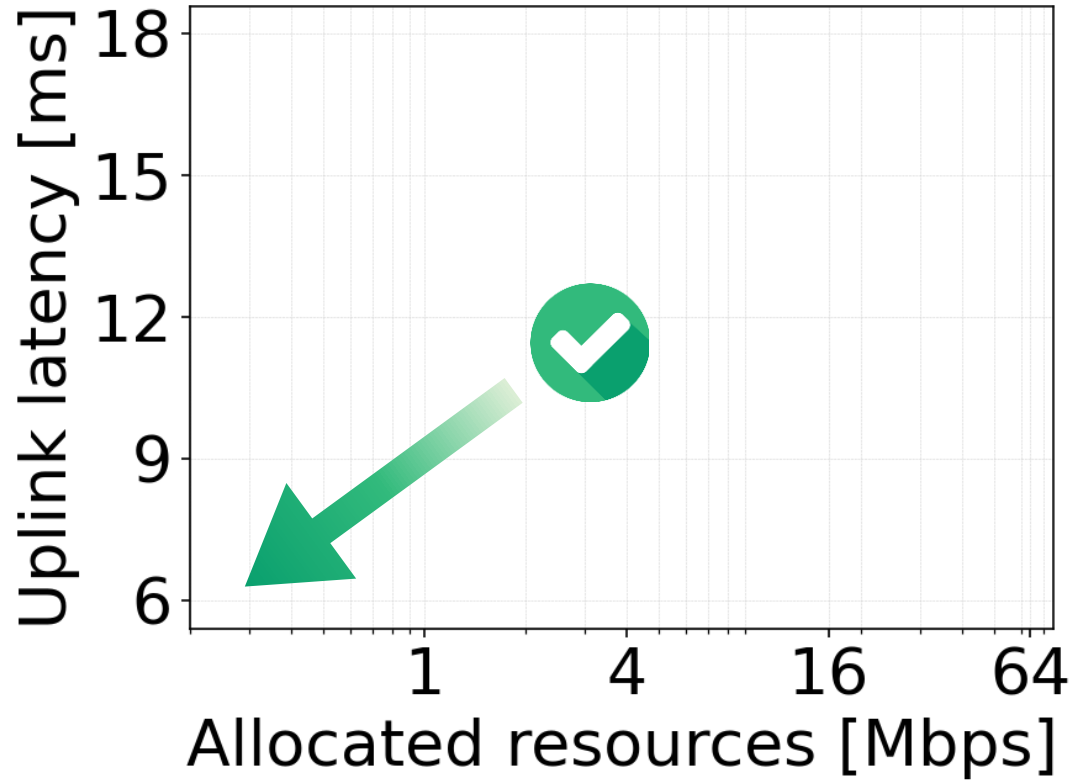
Support runtime updates to the model database

Implementation & Testbed

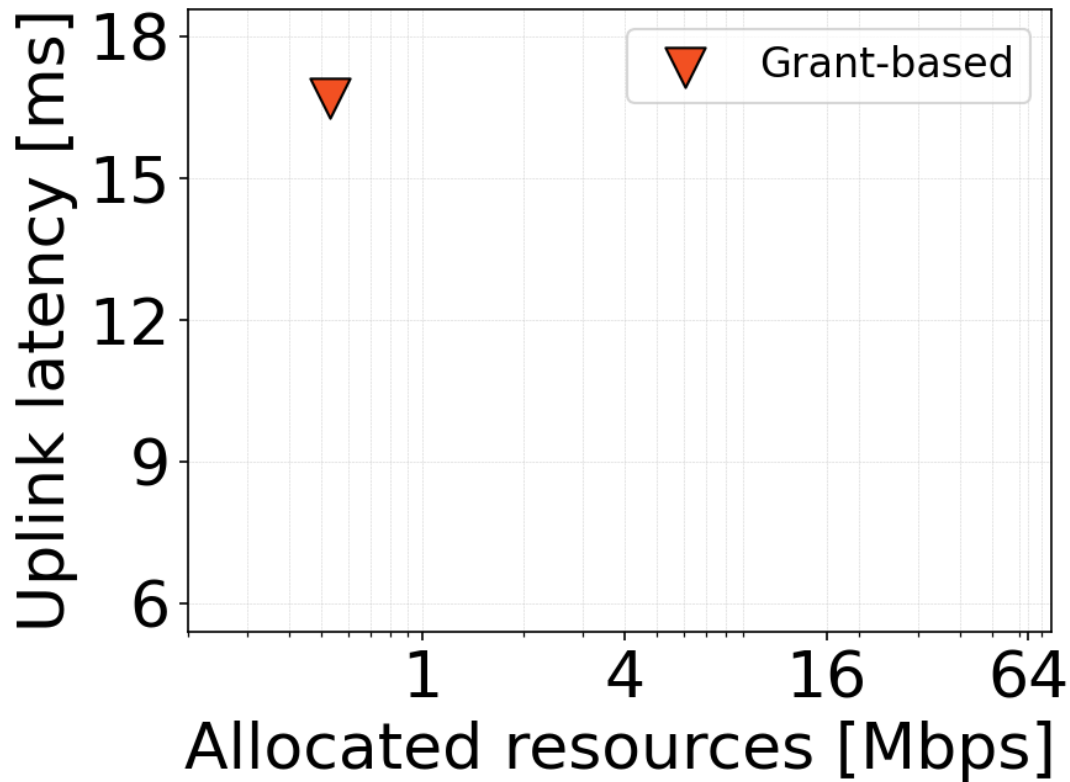


Real-world 5G testbed with commercial 5G users

Evaluation

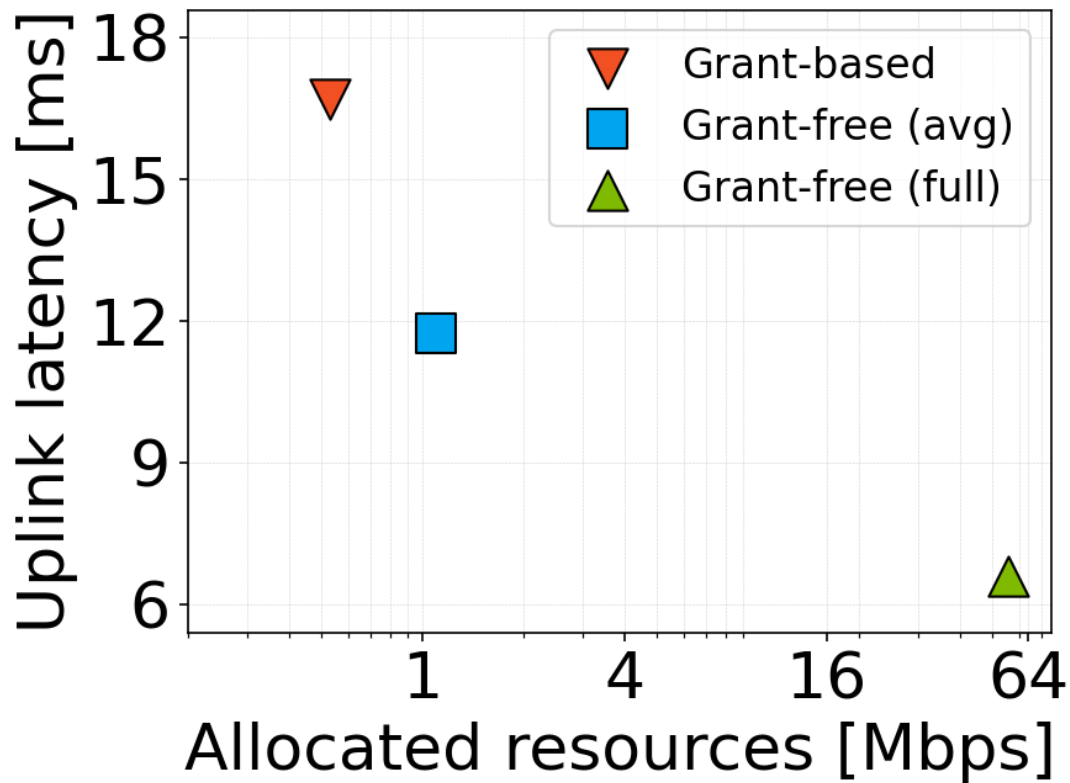


Baselines



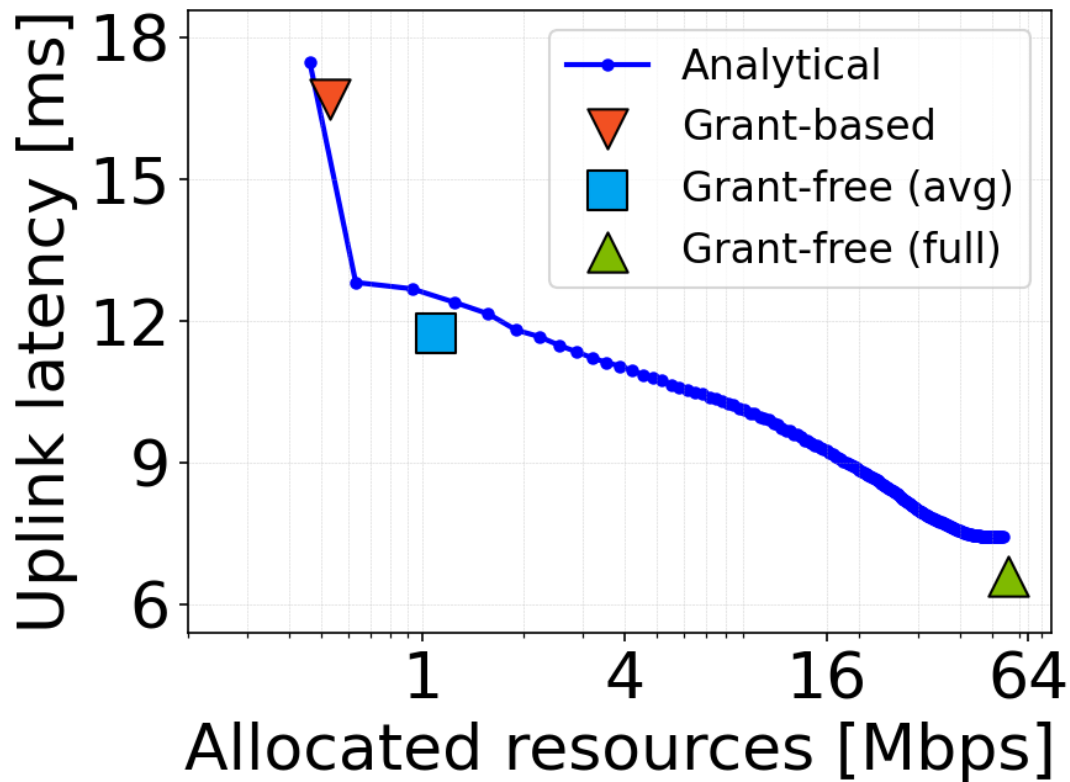
(a) Surveillance cameras

Baselines



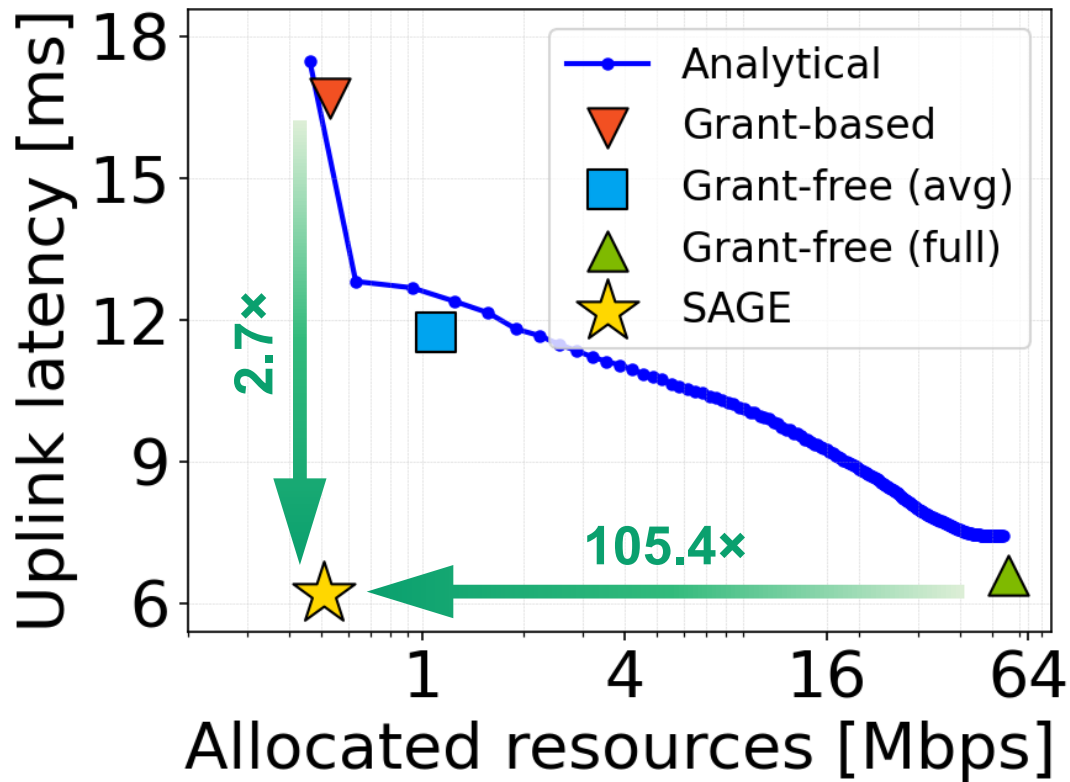
(a) Surveillance cameras

Baselines



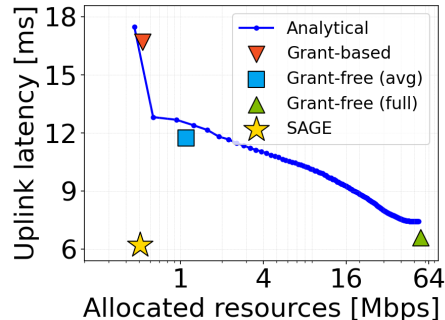
(a) Surveillance cameras

Baselines

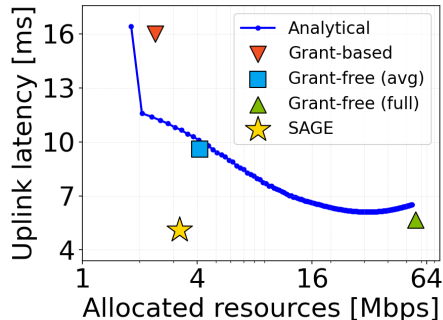


(a) Surveillance cameras

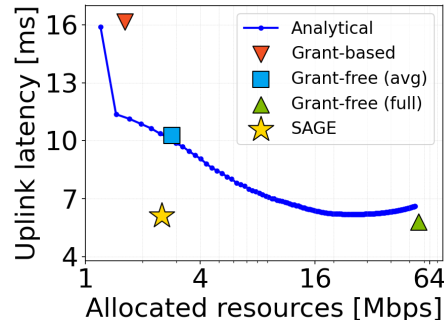
Evaluation Across Different Applications



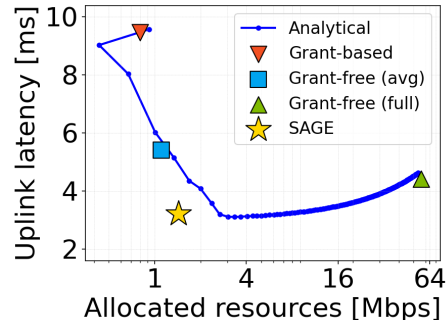
(a) Surveillance cameras



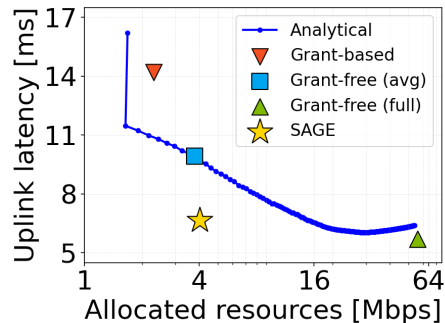
(b) Mobile AR



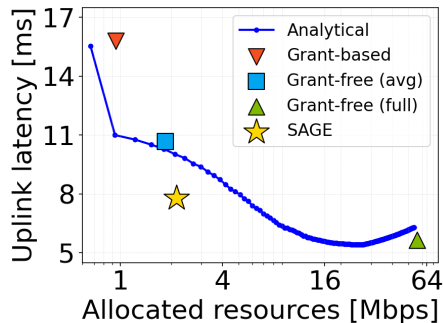
(c) SLAM



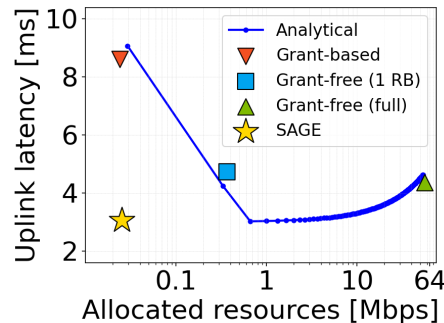
(d) Counter-Strike 2



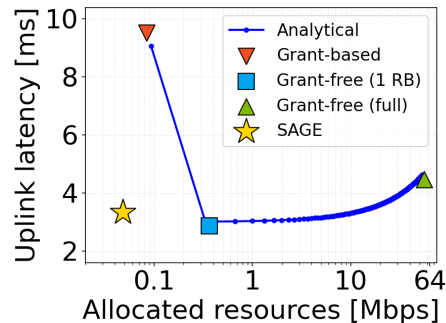
(e) Google Meet



(f) WhatsApp



(g) Industrial automation



(h) SCADA system

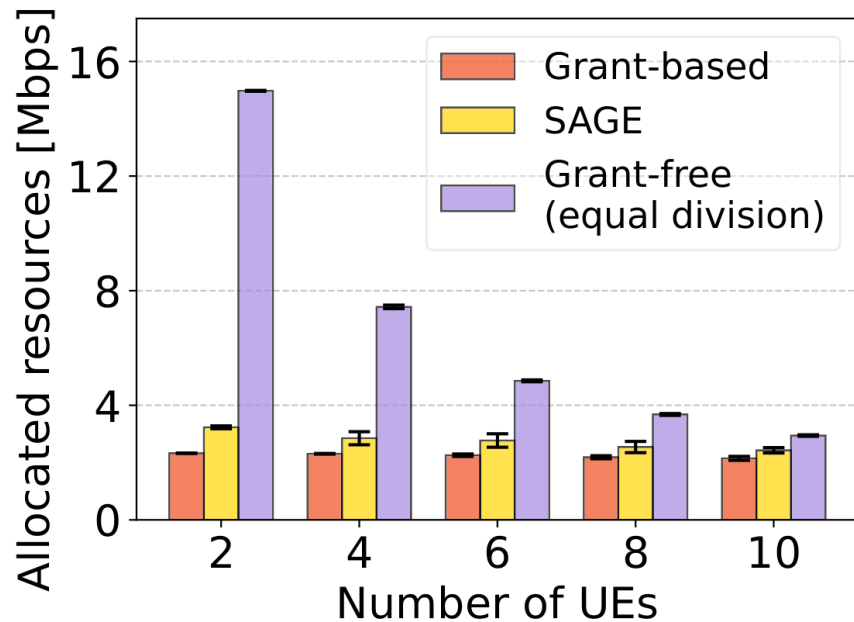
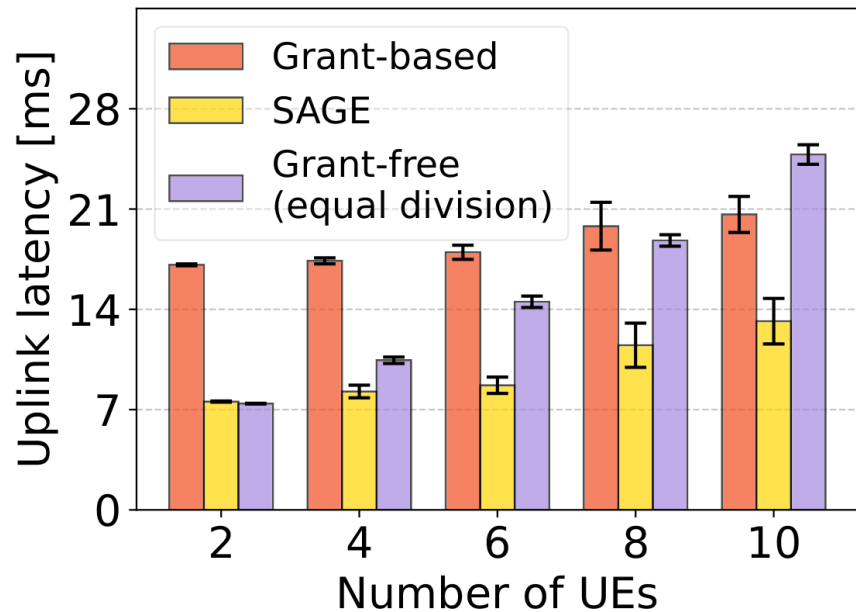
Uplink latency 2.5×



Allocated resources 37.5×

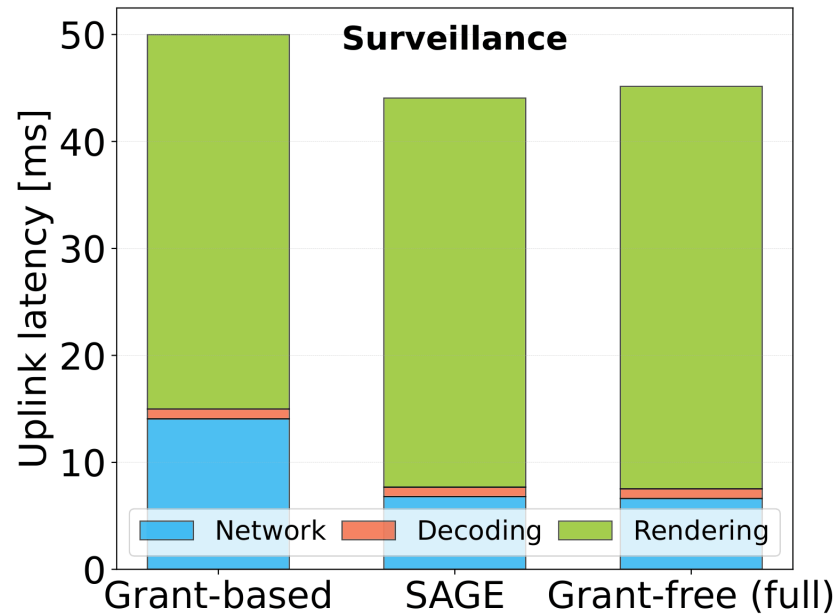
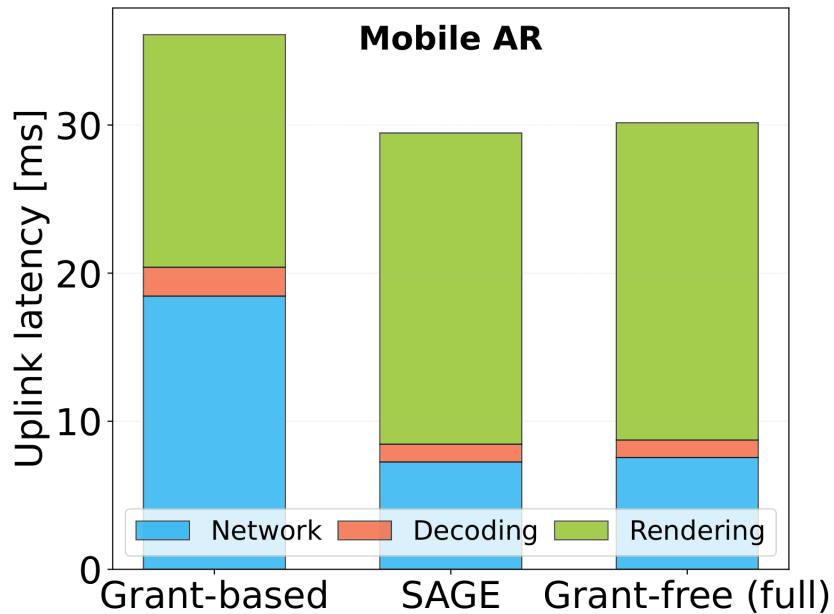


Multi-User Scenario



SAGE scales well under high load and resource contention

Application-Layer Latency



SAGE improves end-to-end application latency

Contributions & Takeaways

A real-time AI system for millisecond-level uplink traffic prediction

A new abstraction traffic trains for reliable and efficient prediction

A full system architecture for real-time deployment

Real-world implementation and extensive over-the-air evaluation



- **What to feed into AI**



- **How to organize and use AI**